WallDot Portal — Flutter Screen Test Catalog

Production-grade, screen-by-screen test cases for the wd_portal_app_flutter application

Residential · Commercial · Interior · Renovation construction scenarios in India, with explicit Flutter implementation gaps

Document	Flutter Screen Test Case Catalog
Target app	wd_portal_app_flutter (Flutter, Dart, GoRouter)
Companion	PORTAL_TEST_CASES.pdf (API-side)
Test cases	241
Flutter gaps documented	47
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Status	Baseline v1.0 — for review with Flutter / QA / PM

1. Scope & Strategy

This catalogue is the Flutter-screen-centric companion to the API test suite. Every case is anchored to a concrete *.dart* file under *lib/screens* or *lib/features*, and the steps are written as a tester actually clicks them — tap this button, swipe this card, observe this widget. Coverage spans all four primary engagement types (Residential, Commercial, Interior, Renovation) plus the secondary Vastu and Smart-Home scopes, with explicit field-ops resilience (Outbox, geotagged photos, offline sync) and India-compliance UI checks.

Roles modelled

Role	Typical screens covered
ADMIN	ACL editor, portal users, partnerships, holidays, WBS templates
DIRECTOR	VO approvals, deduction decisions, Final Account agree
COMMERCIAL_MANAGER / PM	Project detail, Gantt, BOQ doc, CO, stage payments, FA
FINANCE	Invoices, payments, challan, payment schedule
ESTIMATOR / SALES	Leads, lead-estimation wizard, BOQ list
SITE_ENGINEER / SUPERVISOR	Site reports, site visits, mark-complete, attendance
QUALITY_SAFETY (QS)	Quality checks, observations
PROCUREMENT	Vendors, POs, quotations, GRN
CUSTOMER (portal)	Project view, BOQ acknowledge, support tickets
PARTNER (Architect/Consultant)	Referrals, partner dashboard

2. Devices & layouts

Mobile < 850 px — bottom nav (Menu, Projects, Profile) + drawer for SideMenu.
Tablet 850–1100 px — same drawer-based layout in portrait, side-menu in landscape.
Desktop / Web ≥ 1100 px — SideMenu (250 px) + content (flex 5). DPC builder gets a split PDF preview at ≥ 1200 px.
Test devices: Android 11+ (4 GB RAM mid-range plus a flagship), iOS 15+, Chromium / Safari current-2 versions. Network throttling: WiFi, 4G, 3G, offline.

3. Offline-sync model

Mobile and desktop builds use a Drift-backed OutboxService + SyncService stack (S5 PR1/PR2). Submissions of site reports, task mark-complete and delay logs are queued and drained when online. **Web builds skip the entire stack** (F-G30) — they fall back to direct API calls and have no offline capability. Most other screens (labour, inventory, procurement, finance, BOQ, payments) are API-only even on mobile.

4. Self-critique

Where this catalogue stops short:

- Pixel-accurate visual regression (e.g. golden tests) is scoped separately — these cases are functional, not visual.
- Deep Flutter framework concerns (widget rebuild profiling, key uniqueness, RepaintBoundary placement) need devtools work, not a tester click-through.
- Native-platform edge cases (Android scoped storage, iOS background fetch, Web service worker) are listed at high level only; they need platform-specific test rigs.
- Gaps marked F-G## are observed in code today. Several test cases *will fail* because they describe the production-grade UX the app should reach, not what ships now.

5. Screens covered

Screen path	Test cases
lib/screens/auth/portal_login_screen.dart	4
lib/screens/auth/forgot_password_screen.dart	1
lib/screens/auth/reset_password_screen.dart	2
lib/screens/profile/profile_screen.dart	2
lib/screens/main/components/side_menu.dart	3
lib/screens/main/main_screen.dart	3
lib/widgets/offline_banner.dart (used in MainScreen)	1
lib/config/router.dart	1
lib/features/projects/presentation/screens/project_detail_screen.dart	6
lib/responsive.dart (used everywhere)	1
lib/screens/dashboard/crm_dashboard_modern.dart	3
lib/features/leads/presentation/screens/leads_screen.dart	3
lib/features/leads/presentation/screens/add_lead_screen.dart	3
lib/features/leads/presentation/screens/edit_lead_screen.dart	2
lib/features/leads/presentation/screens/lead_activity_screen.dart	1
lib/features/leads/presentation/screens/lead_interactions_screen.dart	1
lib/features/leads/presentation/screens/lead_documents_screen.dart	1
lib/features/lead_estimation/presentation/screens/lead_estimation_wizard_screen.dart	1
lib/features/lead_estimation/presentation/screens/estimation_detail_screen.dart	2
lib/screens/follow_ups/follow_ups_screen.dart	1
lib/screens/communication/communication_screen.dart	1
lib/features/leads/presentation/screens/lead_tasks_screen.dart	1
lib/features/leads/presentation/screens/document_viewer_screen.dart	1
lib/features/customers/presentation/screens/customers_screen.dart	1
lib/features/customers/presentation/screens/add_customer_screen.dart	1
lib/features/customers/presentation/screens/edit_customer_screen.dart	1
lib/features/customers/presentation/screens/customer_detail_screen.dart	3
lib/screens/customer_projects/project_documents_screen.dart	1

lib/screens/customer_projects/project_payments_screen.dart	1
lib/features/projects/presentation/screens/gantt_screen.dart	5
lib/features/projects/presentation/screens/task_progress_entry_screen.dart	3
lib/features/projects/presentation/screens/pm_approval_inbox_screen.dart	2
lib/features/projects/presentation/screens/project_members_screen.dart	1
lib/screens/projects/cctv_camera_form_screen.dart	2
lib/screens/projects/cctv_management_screen.dart	1
lib/screens/projects/subcontracts_screen.dart	1
lib/screens/projects/view_360_player_screen.dart	1
lib/screens/tasks/task_list_screen.dart	1
lib/screens/tasks/task_create_screen.dart	3
lib/screens/tasks/task_alert_dashboard_screen.dart	2
lib/features/scheduling/presentation/screens/wbs_template_editor_screen.dart	2
lib/features/scheduling/presentation/screens/wbs_template_list_screen.dart	1
lib/features/scheduling/presentation/screens/holiday_calendar_screen.dart	2
lib/screens/reports/add_site_report_screen.dart	1
lib/features/site_reports/presentation/screens/site_reports_screen.dart	6
lib/screens/sync/pending_sync_screen.dart + SyncService	1
lib/screens/sync/pending_sync_screen.dart	4
lib/features/site_reports/...	5
Site report detail	1
lib/services/sync_service.dart	4
lib/screens/site_visits/site_visit_check_in_dialog.dart	1
lib/screens/site_visits/site_visits_screen.dart	1
lib/screens/site_visits/site_visit_check_out_dialog.dart	2
lib/screens/site_visits/site_visit_detail_screen.dart	1
lib/screens/site_visits/...	1
lib/features/observations/presentation/screens/observations_screen.dart	1
lib/features/observations/... + task progress screen	1
lib/features/quality/presentation/screens/quality_checks_screen.dart	1
lib/features/gallery/presentation/screens/gallery_screen.dart	1

lib/features/delays/presentation/screens/delay_logs_screen.dart	2
lib/screens/labour/attendance_screen.dart	2
lib/screens/labour/add_labour_screen.dart	1
lib/screens/labour/mb_entry_screen.dart	1
lib/screens/labour/mb_list_screen.dart	1
lib/features/labour/presentation/screens/wage_sheet_screen.dart	1
lib/screens/inventory/materials_screen.dart	1
lib/screens/inventory/add_stock_adjustment_screen.dart	1
lib/screens/inventory/material_consumption_screen.dart	1
lib/screens/inventory/stock_report_screen.dart	1
lib/screens/procurement/add_vendor_screen.dart	2
lib/screens/procurement/po_list_screen.dart	1
lib/screens/procurement/add_purchase_order_screen.dart	1
lib/features/procurement/presentation/screens/quotation_management_screen.dart	2
lib/features/boq/presentation/screens/boq_screen.dart	2
lib/features/boq/presentation/screens/boq_document_management_screen.dart	2
lib/features/boq/presentation/screens/boq_invoice_screen.dart	1
lib/features/boq/presentation/screens/co_management_screen.dart	1
lib/features/boq/presentation/screens/payment_schedule_screen.dart	2
lib/features/boq/...	1
lib/features/dpc/presentation/screens/dpc_builder_screen.dart	5
lib/features/dpc/presentation/screens/dpc_revisions_screen.dart	1
lib/features/dpc_customization_catalog/presentation/screens/dpc_customization_catalog_admin_screen.dart	1
lib/features/estimation_settings/presentation/screens/packages_list_screen.dart	1
lib/features/estimation_settings/presentation/screens/market_index_screen.dart	1
lib/features/estimation_settings/presentation/screens/rate_card_screen.dart	1
lib/screens/finance/finance_dashboard_screen.dart	1
lib/screens/finance/add_project_invoice_screen.dart	1
lib/screens/payments/payments_dashboard_screen.dart	1
lib/screens/payments/challan_management_screen.dart	1
lib/screens/payments/payment_history_screen.dart	1

lib/features/stage_payments/presentation/screens/stage_payment_screen.dart	3
lib/features/deductions/presentation/screens/deduction_register_screen.dart	2
lib/features/final_account/presentation/screens/final_account_screen.dart	3
lib/features/variation_orders/presentation/screens/vo_list_screen.dart	1
lib/features/variation_orders/presentation/screens/vo_create_screen.dart	1
lib/features/variation_orders/presentation/screens/vo_detail_screen.dart	2
lib/screens/notifications/portal_notification_screen.dart	2
lib/screens/support/support_tickets_screen.dart	1
lib/screens/support/support_ticket_detail_screen.dart	1
lib/features/feedback/presentation/screens/feedback_screen.dart	1
lib/features/partnerships/presentation/screens/partnerships_admin_screen.dart	1
lib/features/partnerships/presentation/screens/partner_admin_detail_screen.dart	1
lib/features/warranties/presentation/screens/warranties_screen.dart	2
lib/screens/documents/project_document_list_screen.dart	1
lib/screens/documents/approval_center_screen.dart	2
lib/features/shared/universal_file_viewer_screen.dart	1
lib/screens/documents/design_agreement_screen.dart	1
lib/screens/acl/acl_screen.dart	4
lib/screens/portal_users/add_portal_user_screen.dart	1
lib/screens/portal_users/portal_users_screen.dart	1
lib/screens/team_members/team_members_screen.dart	1
lib/main.dart	4
multiple	8
lib/features/boq/.../payment_schedule_screen.dart	1
lib/features/boq/.../boq_invoice_screen.dart	1
Finance / vendor payment dialog	1
lib/screens/inventory/add_material_screen.dart	1
lib/screens/labour/labour_list_screen.dart	1
Project create / customer projects	1
theme / lib/theme/app_theme.dart	1
lib/theme/app_theme.dart	1

all	2
lib/services/outbox_service.dart	2
lib/services/api_service.dart	1
lib/services/storage_service.dart	1
lib/main.dart + lib/services/push_notification_service.dart	1
lib/services/push_notification_service.dart	3
lib/services/portal_auth_service.dart	1
lib/main.dart + version check	1
customer-facing routes (TBD)	1
customer BOQ document view	1
customer payment screen	1
forms across app	1
any notes field	1
web build	1
lead documents / brochure (BrochureController API)	1
lib/screens/projects/subcontract_work_order_detail_screen.dart	1
lib/screens/boq/boq_line_editor.dart	2
lib/screens/final_account/final_account_editor.dart	1
lib/screens/projects/project_detail_screen.dart	2
lib/screens/customers/customer_list_screen.dart	2
lib/services/outbox_service.dart + Drift schema	1
lib/utils/money_format.dart (or shared)	1
lib/screens/invoice/invoice_preview_screen.dart	1
lib/screens/leads/lead_detail_screen.dart	1
lib/screens/leads/quick_log_interaction_sheet.dart	1
lib/app_shell.dart + offline gates	1
lib/screens/boq/boq_list_screen.dart	1

6. Test cases (by screen cluster)

lib/screens/auth

ID	F-AUTH-001	**Screen**	lib/screens/auth/portal_login_screen.dart
Title	Login with valid credentials on mobile portrait	**Role**	ADMIN / any portal user
Priority	P1	**Type**	Functional
Pre-conditions	Active portal user; mobile in portrait; portal_login_screen open.		
Screen flow / Steps	**1.** Tap email field; enter valid email. **2.** Tap password field; enter valid password (>= 6 chars). **3.** Tap 'Sign In'. **4.** Observe button changes to spinner via isLoading.		
Expected UI behaviour	PortalAuthProvider authenticates. Router redirects to MainScreen / DashboardScreen. Bottom-nav appears with Menu / Projects / Profile (responsive mobile layout).		
Construction context	—	**Flutter gap**	—

ID	F-AUTH-002	**Screen**	lib/screens/auth/portal_login_screen.dart
Title	Login email regex rejects malformed input	**Role**	any
Priority	P2	**Type**	Validation
Pre-conditions	Login screen open.		
Screen flow / Steps	**1.** Enter 'foo@bar' (no TLD) in email. **2.** Enter any password and tap Sign In.		
Expected UI behaviour	Form validator (regex `[\w-.]+@([\w-]+\.)+[\w-]{2,4}`) shows inline error. Shake animation triggers. Field focused.		
Construction context	—	**Flutter gap**	F-G02 — regex pre-2026 RFC; rejects valid TLDs > 4 chars (e.g. .museum).

ID	F-AUTH-003	**Screen**	lib/screens/auth/portal_login_screen.dart
Title	Rapid double-tap on Sign In does not double-submit	**Role**	any
Priority	P2	**Type**	Race / UI
Pre-conditions	Login screen, valid credentials filled.		
Screen flow / Steps	**1.** Tap Sign In twice in < 100 ms.		
Expected UI behaviour	Button disables on first tap; second tap is a no-op. Only one POST /auth/login fires.		
Construction context	—	**Flutter gap**	F-G03 — disable flag races with setState; verify gating logic.

ID	F-AUTH-004	**Screen**	lib/screens/auth/forgot_password_screen.dart
Title	Forgot-password generic 200 response prevents account enumeration	**Role**	any
Priority	P1	**Type**	Security
Pre-conditions	Forgot-password screen open.		
Screen flow / Steps	**1.** Submit a known email — observe success view. **2.** Submit a random unknown email — observe response.		
Expected UI behaviour	Both submits show the same 'Check Your Email' success view. No timing difference, no error banner, no info-leak.		
Construction context	—	**Flutter gap**	—

ID	F-AUTH-005	**Screen**	lib/screens/auth/reset_password_screen.dart
Title	Reset-password token-length validation gate	**Role**	any
Priority	P2	**Type**	Validation
Pre-conditions	Open /reset-password?token=foo (length <= 20).		
Screen flow / Steps	**1.** Wait for build.		
Expected UI behaviour	_buildInvalidTokenView() renders 'broken link' state — UI never shows the password form.		
Construction context	—	**Flutter gap**	F-G02 — token guard is only `length > 20`, not a UUID/sig check.

ID	F-AUTH-006	**Screen**	lib/screens/auth/reset_password_screen.dart
Title	Password mismatch is caught client-side	**Role**	any
Priority	P2	**Type**	Validation
Pre-conditions	Valid reset token in URL.		
Screen flow / Steps	**1.** Enter password 'Abc12345' and confirm 'Abc12346'. **2.** Tap Reset.		
Expected UI behaviour	Confirm-password validator fires 'Passwords do not match'. No API call.		
Construction context	—	**Flutter gap**	—

ID	F-A11Y-001	**Screen**	lib/screens/auth/portal_login_screen.dart
Title	Screen reader on main login flow	**Role**	any (NVDA / TalkBack)
Priority	P3	**Type**	Accessibility
Pre-conditions	Screen reader on.		
Screen flow / Steps	**1.** Navigate through email / password / sign in.		
Expected UI behaviour	Each field announces label + state. Today: limited Semantics.		
Construction context	—	**Flutter gap**	F-G29 — no Semantics widgets added.

lib/screens/profile

ID	F-AUTH-007	**Screen**	lib/screens/profile/profile_screen.dart
Title	Profile screen — change password invalidates current session	**Role**	any logged-in
Priority	P1	**Type**	Security
Pre-conditions	User logged in; profile screen open.		
Screen flow / Steps	**1.** Tap 'Change Password'. **2.** Enter current, new (>= 8 chars), confirm; tap Save.		
Expected UI behaviour	Snackbar success. Per backend G-44 fix all refresh tokens revoked. UI should also force re-login on this device — currently does not.		
Construction context	—	**Flutter gap**	F-G36 — Flutter does not call logout/relogin after password change.

ID	F-AUTH-008	**Screen**	lib/screens/profile/profile_screen.dart
Title	Logout returns to login and clears notification badge	**Role**	any
Priority	P2	**Type**	Functional
Pre-conditions	Logged in; unread notifications present.		
Screen flow / Steps	**1.** Open Profile → tap Logout.		
Expected UI behaviour	GoRouter redirects to /login. Notification badge in MainScreen AppBar disappears. Secure storage cleared.		
Construction context	—	**Flutter gap**	—

lib/screens/main

ID	F-AUTH-009	**Screen**	lib/screens/main/components/side_menu.dart
Title	Avatar initial does not crash on empty firstName	**Role**	any
Priority	P2	**Type**	Edge
Pre-conditions	Backend returns user with empty firstName.		
Screen flow / Steps	**1.** Login; observe side-menu footer & profile screen header.		
Expected UI behaviour	Avatar shows '?' / placeholder. No String.charAt(0) exception.		
Construction context	—	**Flutter gap**	F-G09 — code uses `firstName[0]` without empty guard.

ID	F-NAV-001	**Screen**	lib/screens/main/components/side_menu.dart
Title	Side menu groups expand/collapse with state retained	**Role**	any portal user
Priority	P2	**Type**	UI
Pre-conditions	Desktop (>=1100px) layout; SideMenu visible.		
Screen flow / Steps	**1.** Collapse the CRM group; navigate to a screen and return.		
Expected UI behaviour	Group state persists for the session — collapse flag preserved.		
Construction context	—	**Flutter gap**	F-G17 — verify expansion map is provider-backed, not StatefulWidget local.

ID	F-NAV-002	**Screen**	lib/screens/main/components/side_menu.dart
Title	Menu items hidden when user lacks permission	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Security / RBAC
Pre-conditions	Site engineer login (no FINANCE perms).		
Screen flow / Steps	**1.** Open side menu.		
Expected UI behaviour	Finance, Procurement, ACL items hidden via PermissionProvider.isVisible() callbacks.		
Construction context	—	**Flutter gap**	—

ID	F-NAV-003	**Screen**	lib/screens/main/main_screen.dart
Title	Bottom nav (mobile) — exactly three items map correctly	**Role**	any
Priority	P2	**Type**	Responsive
Pre-conditions	Mobile / tablet (<1100px).		
Screen flow / Steps	**1.** Tap each bottom-nav icon: Menu, Projects, Profile.		
Expected UI behaviour	Menu opens Drawer with SideMenu. Projects routes /projects (index 3). Profile routes /profile (index 15). No flicker, no index drift if items added later.		
Construction context	—	**Flutter gap**	F-G14 — hard-coded indices fragile to menu reordering.

ID	F-NAV-004	**Screen**	lib/screens/main/main_screen.dart
Title	Notification badge — stale value after long background	**Role**	any
Priority	P2	**Type**	Resilience
Pre-conditions	App backgrounded > 30 min with notifications arriving.		
Screen flow / Steps	**1.** Bring app to foreground. **2.** Observe badge count.		
Expected UI behaviour	On AppLifecycleState.resumed, MainScreen should re-fetch unread-count. Today it only fetches in initState and after returning from PortalNotificationScreen.		
Construction context	—	**Flutter gap**	F-G31 — no lifecycle observer for badge refresh.

ID	F-RSP-001	**Screen**	lib/screens/main/main_screen.dart
Title	Side menu vs drawer transitions at 1100px	**Role**	any
Priority	P2	**Type**	Responsive
Pre-conditions	Tablet at 1099px width.		
Screen flow / Steps	**1.** Resize window to 1100px.		
Expected UI behaviour	Layout flips from drawer + bottom nav to sidebar + content. No flicker. Selected menu item preserved.		
Construction context	—	**Flutter gap**	—

lib/widgets/offline_banner.dart (used in MainScreen)

ID	F-NAV-005	**Screen**	lib/widgets/offline_banner.dart (used in MainScreen)
Title	Offline banner appears and links to /sync/pending	**Role**	any
Priority	P1	**Type**	Offline
Pre-conditions	Mobile/desktop; outbox has >=1 pending entry; toggle airplane mode.		
Screen flow / Steps	**1.** Turn airplane mode on. **2.** Observe banner above main content. **3.** Tap banner.		
Expected UI behaviour	Banner: 'Offline — X items queued'. Tap routes to PendingSyncScreen.		
Construction context	—	**Flutter gap**	—

lib/config/router.dart

ID	F-NAV-006	**Screen**	lib/config/router.dart
Title	Deep link to /reset-password?token=… opens reset screen pre-auth	**Role**	anonymous
Priority	P1	**Type**	Routing
Pre-conditions	App cold-start.		
Screen flow / Steps	**1.** Open device deep link wdportal://reset-password?token=.		
Expected UI behaviour	ResetPasswordScreen renders directly with token bound; router redirect logic skips /login for this route.		
Construction context	—	**Flutter gap**	—

lib/features/projects

ID	F-NAV-007	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	ProjectDetailScreen has no deep link	**Role**	PM
Priority	P3	**Type**	UX gap
Pre-conditions	App running.		
Screen flow / Steps	**1.** Try /projects/42 via browser URL or deep link.		
Expected UI behaviour	Today: route not registered — falls back to projects list. Bookmark or share a single project is impossible (except for DPC builder, which uses go_router).		
Construction context	—	**Flutter gap**	F-G03 — no go_router entry for /projects/:id detail.

ID	F-CUST-009	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	Customer project — phase chip edit dialog	**Role**	PM / ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Project in PLANNING.		
Screen flow / Steps	**1.** Tap the phase chip. **2.** Pick DESIGN. **3.** Confirm.		
Expected UI behaviour	SimpleDialog with 5 options. Save calls CustomerProjectService.updateProject. Header re-renders DESIGN chip.		
Construction context	—	**Flutter gap**	F-G15 — no monotonic-transition guard client-side; user could pick PLANNING from CONSTRUCTION (backend should reject per backend G-07 gap).

ID	F-CUST-010	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	GPS lock card — admin-only override visible	**Role**	ADMIN vs PM
Priority	P2	**Type**	Security / RBAC
Pre-conditions	Project with locked GPS.		
Screen flow / Steps	**1.** Open project as PM — note no override button. **2.** Logout, login as ADMIN — observe.		
Expected UI behaviour	canOverride bound to roleCode == 'ADMIN'. PM sees read-only GPS.		
Construction context	—	**Flutter gap**	—

ID	F-PRJ-001	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	Project module grid renders 20+ tiles responsively	**Role**	PM
Priority	P2	**Type**	Responsive
Pre-conditions	Tablet 850px width.		
Screen flow / Steps	**1.** Open project detail.		
Expected UI behaviour	GridView with 2-4 columns (responsive). Tile tap routes to module screen; DPC tile uses context.go('/dpc/builder/{id}'); others use Navigator.push.		
Construction context	—	**Flutter gap**	F-G03 — mixed routing approach inconsistent.

ID	F-PRJ-002	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	Schedule-config tab hidden silently without perms	**Role**	SITE_ENGINEER
Priority	P2	**Type**	UX gap
Pre-conditions	Site engineer (no HOLIDAY_VIEW, no PROJECT_SCHEDULE_CONFIG_EDIT).		
Screen flow / Steps	**1.** Open project detail.		
Expected UI behaviour	Today: schedule-config tab simply not rendered — user has no idea it exists. Better: show greyed-out tab with 'Requires permission X'.		
Construction context	—	**Flutter gap**	F-G10 — silent gating; no Access Denied affordance.

ID	F-GNT-001	**Screen**	lib/features/projects/presentation/screens/gantt_screen.dart
Title	Gantt loads with critical path highlighted red	**Role**	PM
Priority	P1	**Type**	Construction-Domain
Pre-conditions	Project with 30 tasks, predecessors, CPM computed.		
Screen flow / Steps	**1.** Open Project → Gantt.		
Expected UI behaviour	GanttCpmProvider loads; critical tasks rendered in red; float tasks show amber float bar; today line dashed orange.		
Construction context	—	**Flutter gap**	—

ID	F-GNT-002	**Screen**	lib/features/projects/presentation/screens/gantt_screen.dart
Title	Gantt — monsoon-sensitive chip surfaces in monsoon window	**Role**	PM / Site engineer
Priority	P2	**Type**	Construction-Domain
Pre-conditions	Project in Karnataka; monsoon Jun–Sep configured; task scheduled in July.		
Screen flow / Steps	**1.** Open Gantt in current monsoon window.		
Expected UI behaviour	Monsoon icon + warning chip on task label row.		
Construction context	—	**Flutter gap**	F-G12 — flag visible only in Gantt; absent from task list / create screens.

ID	F-GNT-003	**Screen**	lib/features/projects/presentation/screens/gantt_screen.dart
Title	Gantt 100-task performance	**Role**	PM
Priority	P2	**Type**	Performance
Pre-conditions	Project with 150 tasks.		
Screen flow / Steps	**1.** Open Gantt on mid-range Android (4GB RAM). **2.** Scroll the chart laterally and vertically.		
Expected UI behaviour	< 16 ms frame time. Today: ListView.separated may jank > 100 rows.		
Construction context	—	**Flutter gap**	F-G04 — non-virtualised chart row builder.

ID	F-GNT-004	**Screen**	lib/features/projects/presentation/screens/gantt_screen.dart
Title	Inline task date edit dialog — progress slider clamps 0..100	**Role**	PM
Priority	P2	**Type**	Validation
Pre-conditions	Gantt open.		
Screen flow / Steps	**1.** Tap a bar; pick start/end dates; drag progress to 100%; Save.		
Expected UI behaviour	PUT /tasks/{id}/schedule succeeds; bar reflows; CPM recomputes; status auto-derives COMPLETED.		
Construction context	—	**Flutter gap**	—

ID	F-GNT-005	**Screen**	lib/features/projects/presentation/screens/gantt_screen.dart
Title	Predecessor cycle introduced via Gantt edit	**Role**	PM
Priority	P1	**Type**	Construction-Domain
Pre-conditions	A→B→C exists.		
Screen flow / Steps	**1.** Add predecessor C→A via predecessor dialog.		
Expected UI behaviour	Server rejects with CycleDetectedException. UI shows snackbar.		
Construction context	—	**Flutter gap**	F-G05 — no client-side BFS check before submit.

ID	F-TSK-PROG-001	**Screen**	lib/features/projects/presentation/screens/task_progress_entry_screen.dart
Title	Task progress entry — offline-aware mark-complete	**Role**	SITE_ENGINEER (mobile)
Priority	P1	**Type**	Offline
Pre-conditions	Mobile in airplane mode; project assigned.		
Screen flow / Steps	**1.** Open Project → Tasks → Update Progress. **2.** Filter='Active'; tap a task. **3.** Slide progress to 100 %. **4.** Tap 'Mark Complete'. **5.** Grant camera + location.		
Expected UI behaviour	States cycle: idle → capturing → queued. Banner 'Queued — syncs when online'. OutboxService stores SiteReportCreate+TaskMarkComplete.		
Construction context	Residential field-ops day-to-day.	**Flutter gap**	—

ID	F-TSK-PROG-002	**Screen**	lib/features/projects/presentation/screens/task_progress_entry_screen.dart
Title	Progress entry without photo allowed but logged	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Functional
Pre-conditions	Online.		
Screen flow / Steps	**1.** Set progress 60 %; add note; Save.		
Expected UI behaviour	Progress saved without photo. (Photo only mandatory on mark-complete.)		
Construction context	—	**Flutter gap**	F-G39 — distinction between 'update %' (no photo) and 'mark complete' (photo required) easy to miss; needs clearer UX.

ID	F-TSK-PROG-003	**Screen**	lib/features/projects/presentation/screens/task_progress_entry_screen.dart
Title	GPS denied surfaces actionable error state	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Negative
Pre-conditions	Location permission revoked.		
Screen flow / Steps	**1.** Try Mark Complete.		
Expected UI behaviour	State 'errorGpsUnavailable' shown with explanatory text + Retry button.		
Construction context	—	**Flutter gap**	—

ID	F-PM-APP-001	**Screen**	lib/features/projects/presentation/screens/pm_approval_inbox_screen.dart
Title	PM approval inbox — photo thumbnail, accept, reject reason dialog	**Role**	PM (TASK_COMPLETION_APPROVE)
Priority	P1	**Type**	Functional
Pre-conditions	Two queued completions from site engineer (with photos).		
Screen flow / Steps	**1.** Open Approval Inbox. **2.** Tap thumbnail to expand (fullscreen). **3.** Approve first; Reject second with reason 'Plaster not finished'.		
Expected UI behaviour	Approve removes row, completes task. Reject opens RejectCompletionDialog; reason recorded; task reverts to IN_PROGRESS with rejectionReason.		
Construction context	—	**Flutter gap**	F-G13 — only 56 px thumbnail; no fullscreen preview; no bulk approve.

ID	F-PM-APP-002	**Screen**	lib/features/projects/presentation/screens/pm_approval_inbox_screen.dart
Title	Approval inbox blocked for non-PM	**Role**	SITE_ENGINEER
Priority	P1	**Type**	RBAC
Pre-conditions	Site engineer (no approval perm).		
Screen flow / Steps	**1.** Navigate to /tasks/approval-inbox.		
Expected UI behaviour	Screen renders 'You are not authorised…' without fetching data.		
Construction context	—	**Flutter gap**	—

ID	F-PRJ-MEM-001	**Screen**	lib/features/projects/presentation/screens/project_members_screen.dart
Title	Project members — remove secondary signatory with confirm	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Project with primary owner + 2 secondary.		
Screen flow / Steps	**1.** Tap delete on a secondary member; confirm.		
Expected UI behaviour	DELETE call; member removed; owner remains; customer portal access revoked.		
Construction context	—	**Flutter gap**	—

ID	F-MOD-001	**Screen**	lib/features/projects/presentation/screens/project_detail_screen.dart
Title	Module grid 'coming soon' affordance	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	Tile maps to unimplemented module.		
Screen flow / Steps	**1.** Tap that tile.		
Expected UI behaviour	Snackbar 'Coming soon'. Today: silent or generic error.		
Construction context	—	**Flutter gap**	F-G18 — better disable styling for inactive tiles.

lib/responsive.dart (used everywhere)

ID	F-NAV-008	**Screen**	lib/responsive.dart (used everywhere)
Title	Responsive breakpoint switch (rotate device)	**Role**	any
Priority	P2	**Type**	Responsive
Pre-conditions	Tablet at 900px width portrait.		
Screen flow / Steps	**1.** Rotate to landscape (>=1100px).		
Expected UI behaviour	MainScreen switches from Drawer mode to side-by-side SideMenu + content. AppBar hamburger disappears.		
Construction context	—	**Flutter gap**	—

lib/screens/dashboard

ID	F-DSH-001	**Screen**	lib/screens/dashboard/crm_dashboard_modern.dart
Title	Dashboard skeleton vs. data render	**Role**	ADMIN / PM
Priority	P2	**Type**	Functional
Pre-conditions	Authenticated.		
Screen flow / Steps	**1.** Open dashboard cold.		
Expected UI behaviour	Shimmer skeleton (4 cards × 120px) shows then resolves to KPI tiles. 'Updated X mins ago' label appears bottom.		
Construction context	—	**Flutter gap**	F-G18 — skeleton hard-coded to 4 cards regardless of API payload.

ID	F-DSH-002	**Screen**	lib/screens/dashboard/crm_dashboard_modern.dart
Title	Pull-to-refresh updates timestamps	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	Dashboard open for >5 mins.		
Screen flow / Steps	**1.** Pull down on the scrollable area.		
Expected UI behaviour	RefreshIndicator runs _loadData(); _lastUpdated resets to now.		
Construction context	—	**Flutter gap**	—

ID	F-DSH-003	**Screen**	lib/screens/dashboard/crm_dashboard_modern.dart
Title	Granular error per chart not supported	**Role**	any
Priority	P3	**Type**	Resilience gap
Pre-conditions	One chart endpoint returns 500.		
Screen flow / Steps	**1.** Reload dashboard.		
Expected UI behaviour	Entire dashboard shows a single error widget. Chart-level fallback (other charts succeed) not implemented.		
Construction context	—	**Flutter gap**	F-G18 — single error state for compound dashboard.

lib/features/leads

ID	F-LEAD-001	**Screen**	lib/features/leads/presentation/screens/leads_screen.dart
Title	Lead list pagination via scroll-to-load	**Role**	SALES / ADMIN
Priority	P1	**Type**	Functional
Pre-conditions	50+ leads in backend.		
Screen flow / Steps	**1.** Open Leads screen. **2.** Scroll list; observe 'Loading more…' indicator at 200 px from end.		
Expected UI behaviour	Pages fetched at threshold; no duplicate fetch (loadingMore flag set). Scroll position preserved when items appended.		
Construction context	—	**Flutter gap**	F-G19 — no debounce guard; rapid scroll can re-trigger.

ID	F-LEAD-002	**Screen**	lib/features/leads/presentation/screens/leads_screen.dart
Title	Filter chips combine with search across pagination	**Role**	SALES
Priority	P1	**Type**	Functional
Pre-conditions	Mixed-status leads exist.		
Screen flow / Steps	**1.** Type 'Anita' in search. **2.** Select status='qualified'. **3.** Select projectType='residential_construction'.		
Expected UI behaviour	Search + filters compose into a single PaginationParams; result set drops to matching leads only; page resets to 0.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-003	**Screen**	lib/features/leads/presentation/screens/add_lead_screen.dart
Title	Add Lead — public/walk-in (no email)	**Role**	SALES
Priority	P1	**Type**	Functional
Pre-conditions	Add Lead form open.		
Screen flow / Steps	**1.** Fill name, phone, projectType='residential_construction', state='Karnataka', district='Bengaluru', budget=4500000. **2.** Leave email blank. **3.** Set source='directWalkin'. **4.** Save.		
Expected UI behaviour	Lead created; row appears in list with COLD/WARM/HOT badge computed from scoring factors. Welcome email NOT triggered (no email).		
Construction context	Residential walk-in customer.	**Flutter gap**	—

ID	F-LEAD-004	**Screen**	lib/features/leads/presentation/screens/edit_lead_screen.dart
Title	Lead score recompute on budget edit	**Role**	SALES
Priority	P2	**Type**	Integration
Pre-conditions	Lead with budget < 10L (LOW factor).		
Screen flow / Steps	**1.** Edit budget to 60,00,000. **2.** Save. **3.** Open Lead Score History tab.		
Expected UI behaviour	Score category transitions COLD/WARM → HOT; LeadScoreHistory row visible with previous/new + JSON factors.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-005	**Screen**	lib/features/leads/presentation/screens/edit_lead_screen.dart
Title	Architect referral banner shows on edit	**Role**	SALES
Priority	P2	**Type**	UX
Pre-conditions	Lead.leadSource = 'referralArchitect' with linked partner.		
Screen flow / Steps	**1.** Open Edit Lead.		
Expected UI behaviour	Purple banner with partner firm + phone fetched via PartnershipAdminService. 'Loading partner info…' shows briefly.		
Construction context	—	**Flutter gap**	F-G18 — partner fetched only on initial build; not refreshed when user changes source dropdown mid-edit.

ID	F-LEAD-006	**Screen**	lib/features/leads/presentation/screens/lead_activity_screen.dart
Title	Lead Activity timeline merges interactions + system events	**Role**	SALES / PM
Priority	P2	**Type**	Functional
Pre-conditions	Lead with 3 interactions, 1 score change, 1 doc upload.		
Screen flow / Steps	**1.** Open Activity tab.		
Expected UI behaviour	Timeline shows mixed events in reverse chronological order; icons differ by event type.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-007	**Screen**	lib/features/leads/presentation/screens/lead_interactions_screen.dart
Title	Quick-log interaction schedules next follow-up	**Role**	SALES
Priority	P1	**Type**	Functional
Pre-conditions	Lead in 'qualified'.		
Screen flow / Steps	**1.** Tap '+'; choose 'Quick Log'. **2.** Type='CALL', outcome='SCHEDULED_FOLLOWUP', nextActionDate=+3d. **3.** Save.		
Expected UI behaviour	Card appears top of list; nextFollowUp persisted; FollowUpsScreen 'Overdue' tab will surface on D+3.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-008	**Screen**	lib/features/leads/presentation/screens/lead_documents_screen.dart
Title	Document upload restricts to whitelisted extensions	**Role**	SALES
Priority	P1	**Type**	Validation
Pre-conditions	Lead documents tab open.		
Screen flow / Steps	**1.** Upload a .exe file via file picker. **2.** Upload a 2 MB JPG.		
Expected UI behaviour	.exe rejected with helpful message (whitelist: pdf, jpg, jpeg, png, webp, doc, docx, xls, xlsx, ppt, pptx, csv, txt). JPG succeeds.		
Construction context	—	**Flutter gap**	F-G34 — no upload progress shown; no virus-scan client-side.

ID	F-LEAD-014	**Screen**	lib/features/leads/presentation/screens/lead_tasks_screen.dart
Title	Lead tasks tab loading + delete confirm	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Lead with 5 tasks.		
Screen flow / Steps	**1.** Open Tasks tab. **2.** Tap delete on a task.		
Expected UI behaviour	Loading spinner → tasks sorted by dueDate asc. Delete shows confirmation dialog; on confirm list reloads (no optimistic UI).		
Construction context	—	**Flutter gap**	F-G18 — no optimistic UI on update/delete.

ID	F-LEAD-015	**Screen**	lib/features/leads/presentation/screens/document_viewer_screen.dart
Title	Document viewer renders PDF + image with auth header	**Role**	any
Priority	P2	**Type**	Functional
Pre-conditions	Lead with 1 PDF + 1 JPG document.		
Screen flow / Steps	**1.** Open PDF; open JPG.		
Expected UI behaviour	SfPdfViewer (memory/network) renders PDF. Image opens in InteractiveViewer (pinch zoom 0.5x–4x).		
Construction context	—	**Flutter gap**	F-G24 — no print/share/annotate; no video preview.

ID	F-RES-002	**Screen**	lib/features/leads/presentation/screens/leads_screen.dart
Title	No internet — list shows actionable error state	**Role**	any
Priority	P2	**Type**	Resilience
Pre-conditions	Airplane mode.		
Screen flow / Steps	**1.** Open Leads.		
Expected UI behaviour	ErrorStateWidget with retry button; offline banner above.		
Construction context	—	**Flutter gap**	—

ID	F-KBD-001	**Screen**	lib/features/leads/presentation/screens/add_lead_screen.dart
Title	Soft keyboard does not occlude submit button on long form	**Role**	any
Priority	P2	**Type**	UX
Pre-conditions	Mobile portrait.		
Screen flow / Steps	**1.** Tap on Notes (last field) — keyboard rises. **2.** Scroll to Save button.		
Expected UI behaviour	Scaffold resizes; Save button reachable above keyboard.		
Construction context	—	**Flutter gap**	F-G18 — confirm resizeToAvoidBottomInset / SingleChildScrollView.

ID	F-LIFE-002	**Screen**	lib/features/leads/presentation/screens/add_lead_screen.dart
Title	Phone rotation mid-form preserves field values	**Role**	any
Priority	P2	**Type**	Resilience
Pre-conditions	Form partially filled.		
Screen flow / Steps	**1.** Rotate device to landscape.		
Expected UI behaviour	Field values preserved (Form state retained); no data loss.		
Construction context	—	**Flutter gap**	—

lib/features/lead_estimation

ID	F-LEAD-009	**Screen**	lib/features/lead_estimation/presentation/screens/lead_estimation_wizard_screen.dart
Title	Estimation wizard — budgetary mode with confidence	**Role**	ESTIMATOR / SALES
Priority	P1	**Type**	Construction-Type
Pre-conditions	Lead with projectType='interior_work'.		
Screen flow / Steps	**1.** Open 'Generate Estimation' from Lead Quotations. **2.** Step 1: choose package 'Interior Premium'. **3.** Step 2: switch to budgetary mode; enter estimatedArea=1200, confidenceLevel='Medium'. **4.** Step 3-4: skip (catalog endpoints stub). **5.** Step 5: enter discount 5%, GST 18%, validUntil=+30d. Save.		
Expected UI behaviour	WizardDraft sent to LeadEstimationsProvider; preview JSON computed; estimation row created and visible in lead's quotations list.		
Construction context	Interior — design-only catalog, no structural items.	**Flutter gap**	—

ID	F-LEAD-010	**Screen**	lib/features/lead_estimation/presentation/screens/estimation_detail_screen.dart
Title	Estimation detail — revise opens wizard with pre-fill	**Role**	ESTIMATOR
Priority	P2	**Type**	Functional
Pre-conditions	Estimation in DRAFT or SENT status.		
Screen flow / Steps	**1.** Open estimation detail. **2.** Tap 'Revise'.		
Expected UI behaviour	LeadEstimationWizardScreen opens with existing values bound to WizardDraft. Save creates a new revision.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-011	**Screen**	lib/features/lead_estimation/presentation/screens/estimation_detail_screen.dart
Title	Estimation PDF download shows action spinner	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	Estimation in ISSUED.		
Screen flow / Steps	**1.** Tap PDF download icon.		
Expected UI behaviour	AppBar action shows CircularProgressIndicator until FileDownloadHelper completes.		
Construction context	—	**Flutter gap**	—

lib/screens/follow_ups

ID	F-LEAD-012	**Screen**	lib/screens/follow_ups/follow_ups_screen.dart
Title	Follow-ups screen — overdue badge accuracy	**Role**	SALES
Priority	P2	**Type**	Functional
Pre-conditions	Two leads with nextFollowUp = yesterday.		
Screen flow / Steps	**1.** Open Follow-ups → Overdue tab.		
Expected UI behaviour	Both leads listed; overdue chip red; tap routes to lead detail.		
Construction context	—	**Flutter gap**	F-G19 — type counts computed from current page only.

lib/screens/communication

ID	F-LEAD-013	**Screen**	lib/screens/communication/communication_screen.dart
Title	Communication screen — global interaction search debounce	**Role**	any
Priority	P3	**Type**	Performance
Pre-conditions	1000+ interactions in DB.		
Screen flow / Steps	**1.** Type 'anita' rapidly in search.		
Expected UI behaviour	Today: every keystroke triggers a reload (no debounce). Goal: 300ms debounce; cancel in-flight request.		
Construction context	—	**Flutter gap**	F-G19 — search not debounced.

lib/features/customers

ID	F-CUST-001	**Screen**	lib/features/customers/presentation/screens/customers_screen.dart
Title	Customer list filter chips and search	**Role**	SALES / PM
Priority	P2	**Type**	Functional
Pre-conditions	Mixed individual/business customers seeded.		
Screen flow / Steps	**1.** Tap 'Business' chip. **2.** Type 'Acme'.		
Expected UI behaviour	List filtered to business customers whose name contains 'Acme'. Pagination footer updates totalPages.		
Construction context	—	**Flutter gap**	—

ID	F-CUST-002	**Screen**	lib/features/customers/presentation/screens/add_customer_screen.dart
Title	Add Customer — GST number validation	**Role**	SALES
Priority	P1	**Type**	Validation / India
Pre-conditions	Add Customer form open.		
Screen flow / Steps	**1.** Enter GSTIN '29ABCDE1234F1Z5'. **2.** Enter GSTIN '29ABCDE1234F1Z' (14 chars). **3.** Enter GSTIN '00ABCDE1234F1Z5' (invalid state 00).		
Expected UI behaviour	First accepted. Second rejected (length). Third rejected (state-code check).		
Construction context	—	**Flutter gap**	F-G06 — GSTIN format only regex-checked; no checksum.

ID	F-CUST-003	**Screen**	lib/features/customers/presentation/screens/edit_customer_screen.dart
Title	Edit Customer — password blank means 'keep current'	**Role**	ADMIN
Priority	P2	**Type**	UX
Pre-conditions	Customer exists with password.		
Screen flow / Steps	**1.** Open Edit; leave password blank; change phone. **2.** Save.		
Expected UI behaviour	Phone updated; customer can still login with previous password.		
Construction context	—	**Flutter gap**	F-G18 — placeholder text could mislead user; needs explicit 'leave blank to keep current' helper text.

ID	F-CUST-004	**Screen**	lib/features/customers/presentation/screens/customer_detail_screen.dart
Title	Customer detail — soft-deactivate vs strict delete distinguishable	**Role**	ADMIN
Priority	P1	**Type**	Functional
Pre-conditions	Customer linked to active project.		
Screen flow / Steps	**1.** Open detail. **2.** Tap 'Deactivate' (enabled toggle off). **3.** Then tap 'Delete'.		
Expected UI behaviour	Deactivate succeeds (PATCH /enabled). Delete refused with 409 listing active references; UI shows reason.		
Construction context	—	**Flutter gap**	—

ID	F-CUST-005	**Screen**	lib/features/customers/presentation/screens/customer_detail_screen.dart
Title	Reset password email — single-click guard	**Role**	ADMIN
Priority	P2	**Type**	Race
Pre-conditions	Customer with email.		
Screen flow / Steps	**1.** Tap 'Send Password Reset' twice rapidly.		
Expected UI behaviour	_isSendingReset disables button; only one email sent.		
Construction context	—	**Flutter gap**	—

ID	F-CUST-008	**Screen**	lib/features/customers/presentation/screens/customer_detail_screen.dart
Title	Customer detail — initial-customer fast render	**Role**	any
Priority	P3	**Type**	Performance / UX
Pre-conditions	Coming from list with cached row.		
Screen flow / Steps	**1.** Tap a customer row.		
Expected UI behaviour	Detail renders immediately from initialCustomer; background fetch refreshes; refresh failure silent.		
Construction context	—	**Flutter gap**	F-G18 — silent failure on background refresh; user unaware if data is stale.

lib/screens/customer_projects

ID	F-CUST-006	**Screen**	lib/screens/customer_projects/project_documents_screen.dart
Title	Customer project documents — category filter	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Project with docs in 3 categories.		
Screen flow / Steps	**1.** Open project documents. **2.** Tap 'Permits' chip.		
Expected UI behaviour	List filters to permit docs; chip horizontal scroller does NOT auto-scroll to selected (UX issue).		
Construction context	—	**Flutter gap**	F-G17 — selected chip not auto-scrolled into view.

ID	F-CUST-007	**Screen**	lib/screens/customer_projects/project_payments_screen.dart
Title	Project payments — design payment 'no record' state	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Brand-new project with no design payment yet.		
Screen flow / Steps	**1.** Open project payments.		
Expected UI behaviour	Empty state icon + 'No payment records' (backend returns 200 + data:null per recent change).		
Construction context	—	**Flutter gap**	—

lib/screens/projects

ID	F-CCTV-001	**Screen**	lib/screens/projects/cctv_camera_form_screen.dart
Title	Add CCTV camera form requires name; validates port	**Role**	ADMIN
Priority	P2	**Type**	Validation
Pre-conditions	Add Camera open.		
Screen flow / Steps	**1.** Save with empty name → expect inline error. **2.** Enter name; set port='8080a' → expect parse error. **3.** Fix port=8080; Save.		
Expected UI behaviour	First save blocked. Second blocked by int.tryParse null. Third succeeds.		
Construction context	—	**Flutter gap**	F-G09 — password field stored plaintext in form; no auth test or encryption pre-save.

ID	F-CCTV-002	**Screen**	lib/screens/projects/cctv_management_screen.dart
Title	CCTV management — enable/disable toggle	**Role**	ADMIN
Priority	P3	**Type**	Functional
Pre-conditions	Camera 'Site-Gate' ACTIVE.		
Screen flow / Steps	**1.** Open camera; PopupMenu → Disable.		
Expected UI behaviour	PATCH /cctv-cameras/{id}/toggle; row color changes to grey.		
Construction context	—	**Flutter gap**	—

ID	F-SUB-001	**Screen**	lib/screens/projects/subcontracts_screen.dart
Title	Subcontract list — status filter chips	**Role**	PM / Procurement
Priority	P2	**Type**	Functional
Pre-conditions	Subcontracts in DRAFT/ACTIVE/COMPLETED.		
Screen flow / Steps	**1.** Open Subcontracts. **2.** Filter='Active'.		
Expected UI behaviour	Only active rows. Tapping a row shows snackbar 'View details…' — detail screen not implemented.		
Construction context	—	**Flutter gap**	F-G11 — detail screen stub; F-G38 — retention not shown per row.

ID	F-360-001	**Screen**	lib/screens/projects/view_360_player_screen.dart
Title	View 360 player loads panorama + info sheet	**Role**	any project member
Priority	P3	**Type**	Functional
Pre-conditions	Project with uploaded 360 image.		
Screen flow / Steps	**1.** Open 360 list; tap a tour.		
Expected UI behaviour	Panorama widget; drag-to-explore hint visible 3 s then fades.		
Construction context	—	**Flutter gap**	F-G25 — no VR mode; pinch-zoom dependent on `panorama` package internal handling.

ID	F-SEC-001	**Screen**	lib/screens/projects/cctv_camera_form_screen.dart
Title	CCTV password masked while typing	**Role**	ADMIN
Priority	P1	**Type**	Security
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Type password.		
Expected UI behaviour	obscureText = true; toggle eye to reveal.		
Construction context	—	**Flutter gap**	F-G09 — verify masked; payload not logged in console.

ID	F-RET-001	**Screen**	lib/screens/projects/subcontract_work_order_detail_screen.dart
Title	Subcontract retention release detail visible	**Role**	PM / Finance
Priority	P2	**Type**	Construction-Domain
Pre-conditions	Subcontract with totalRetentionAccumulated = ■50,000.		
Screen flow / Steps	**1.** Open work order detail.		
Expected UI behaviour	Retention card: accumulated, released, balance. Release button gated by DLP end date.		
Construction context	—	**Flutter gap**	F-G38 — retention breakdown only in detail, not list.

ID	F-PRJ-DEL-001	**Screen**	lib/screens/projects/project_detail_screen.dart
Title	Admin-only Delete Project — typed-confirmation dialog	**Role**	ADMIN
Priority	P1	**Type**	Security
Pre-conditions	Project P-77 detail open; viewer is ADMIN.		
Screen flow / Steps	**1.** Tap kebab → Delete Project.		
Expected UI behaviour	Dialog: 'Type project code WD-77 to confirm'. Delete disabled until exact match. After delete, route pops to project list; toast confirms; project hidden from list.		
Construction context	—	**Flutter gap**	—

ID	F-PRJ-DEL-002	**Screen**	lib/screens/projects/project_detail_screen.dart
Title	Non-admin does not see Delete Project entry	**Role**	PROJECT_MANAGER
Priority	P1	**Type**	RBAC
Pre-conditions	Same project, role=PM.		
Screen flow / Steps	**1.** Open kebab.		
Expected UI behaviour	Delete Project not rendered (not just disabled). UI does not leak the option's existence.		
Construction context	—	**Flutter gap**	—

lib/screens/tasks

ID	F-TLST-001	**Screen**	lib/screens/tasks/task_list_screen.dart
Title	Task list status + priority multi-filter	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Mixed tasks seeded.		
Screen flow / Steps	**1.** Tap status='IN_PROGRESS' + priority='HIGH'.		
Expected UI behaviour	List intersect; pagination footer updates.		
Construction context	—	**Flutter gap**	—

ID	F-TCR-001	**Screen**	lib/screens/tasks/task_create_screen.dart
Title	Task create requires dueDate	**Role**	PM
Priority	P1	**Type**	Validation
Pre-conditions	Create form open.		
Screen flow / Steps	**1.** Fill title; leave dueDate empty. **2.** Submit.		
Expected UI behaviour	Form validation blocks. Pick a date; submit succeeds.		
Construction context	—	**Flutter gap**	—

ID	F-TCR-002	**Screen**	lib/screens/tasks/task_create_screen.dart
Title	Date picker honours Indian DD/MM/YYYY	**Role**	any
Priority	P3	**Type**	Localisation
Pre-conditions	Device locale en_IN.		
Screen flow / Steps	**1.** Open due date picker.		
Expected UI behaviour	Picker order is DD/MM/YYYY. Code uses `{day}/{month}/{year}` literal which works in India but not locale-aware.		
Construction context	—	**Flutter gap**	F-G16 — use DateFormat('dd/MM/yyyy', 'en_IN').

ID	F-TCR-003	**Screen**	lib/screens/tasks/task_create_screen.dart
Title	Predecessor selection NOT available on create	**Role**	PM
Priority	P2	**Type**	Gap
Pre-conditions	Create form.		
Screen flow / Steps	**1.** Scan available fields.		
Expected UI behaviour	Today: predecessors not editable on task create — must be added later via WBS template editor or Gantt. Worth surfacing on create.		
Construction context	—	**Flutter gap**	F-G14 — predecessor editor missing on create screen.

ID	F-TALR-001	**Screen**	lib/screens/tasks/task_alert_dashboard_screen.dart
Title	Task alert dashboard — admin manual trigger	**Role**	ADMIN
Priority	P3	**Type**	Functional
Pre-conditions	Admin logged in.		
Screen flow / Steps	**1.** Open Alerts. **2.** Tap 'Trigger Manual Check' icon.		
Expected UI behaviour	POST /tasks/alerts/trigger; recent alerts list refreshes.		
Construction context	—	**Flutter gap**	—

ID	F-TALR-002	**Screen**	lib/screens/tasks/task_alert_dashboard_screen.dart
Title	Severity pie chart renders zero state	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	No alerts.		
Screen flow / Steps	**1.** Open Alerts.		
Expected UI behaviour	Pie chart degraded to placeholder; stat cards show 0/0/0.		
Construction context	—	**Flutter gap**	—

lib/features/scheduling

ID	F-WBS-001	**Screen**	lib/features/scheduling/presentation/screens/wbs_template_editor_screen.dart
Title	WBS template editor — add phase, then task	**Role**	ADMIN (canManageWbsTemplates)
Priority	P1	**Type**	Functional
Pre-conditions	Empty template.		
Screen flow / Steps	**1.** Add phase 'Foundation' via FAB. **2.** Add task 'Excavation', duration 5 days, weight=null, monsoonSensitive=true. **3.** Save.		
Expected UI behaviour	ReorderableListView on phase pane; task appears in DataTable; save returns new template version.		
Construction context	Residential foundation phase.	**Flutter gap**	—

ID	F-WBS-002	**Screen**	lib/features/scheduling/presentation/screens/wbs_template_editor_screen.dart
Title	Predecessor editor missing — only count shown	**Role**	ADMIN
Priority	P2	**Type**	Gap
Pre-conditions	Task added.		
Screen flow / Steps	**1.** Try to edit predecessors from the task row.		
Expected UI behaviour	Today: 'Preds' column shows count only; no editor dialog.		
Construction context	—	**Flutter gap**	F-G14 — predecessor editor not implemented in screen.

ID	F-WBS-003	**Screen**	lib/features/scheduling/presentation/screens/wbs_template_list_screen.dart
Title	Apply WBS template to a project	**Role**	ADMIN / PM
Priority	P1	**Type**	Construction-Type
Pre-conditions	Templates seeded per project type.		
Screen flow / Steps	**1.** Pick template 'Commercial-Standard-G+4'. **2.** Apply to a fresh commercial project.		
Expected UI behaviour	Tasks + predecessors cloned. Dates shifted from project start. Gantt populated.		
Construction context	Commercial G+4.	**Flutter gap**	—

ID	F-HOL-001	**Screen**	lib/features/scheduling/presentation/screens/holiday_calendar_screen.dart
Title	Holiday calendar — scope dropdown required	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Calendar open.		
Screen flow / Steps	**1.** Try to create a holiday without selecting scope.		
Expected UI behaviour	Backend rejects (scope required). UI surfaces error toast.		
Construction context	—	**Flutter gap**	—

ID	F-HOL-002	**Screen**	lib/features/scheduling/presentation/screens/holiday_calendar_screen.dart
Title	State-level holiday overrides global on project Gantt	**Role**	ADMIN
Priority	P3	**Type**	Construction-Domain
Pre-conditions	Project in Karnataka; 'Kannada Rajyotsava' on Nov 1 state-scope.		
Screen flow / Steps	**1.** Schedule a task across Nov 1. **2.** Open Gantt.		
Expected UI behaviour	Nov 1 marked non-working; bar reflows skipping Nov 1.		
Construction context	—	**Flutter gap**	F-G15 — no project-specific override field; cannot disable a state holiday for one project.

lib/screens/reports

ID	F-SR-001	**Screen**	lib/screens/reports/add_site_report_screen.dart
Title	Site report — at-least-one-photo enforced	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Validation
Pre-conditions	Add Site Report open.		
Screen flow / Steps	**1.** Fill title + description; no photos. **2.** Submit.		
Expected UI behaviour	Toast 'At least one photo required'. Form does not submit.		
Construction context	—	**Flutter gap**	—

lib/features/site_reports

ID	F-SR-002	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Photo capture preserves geotag + accuracy	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Construction-Domain
Pre-conditions	On-site; GPS enabled.		
Screen flow / Steps	**1.** Take 3 photos via camera; observe lat/lng/accuracy chip.		
Expected UI behaviour	ImagePicker captures with imageQuality=70; lat/lng/accuracy attached to each PhotoCapture; distanceFromProject computed.		
Construction context	—	**Flutter gap**	—

ID	F-SR-003	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Offline submit queues to outbox with idempotency key	**Role**	SITE_ENGINEER (mobile)
Priority	P1	**Type**	Offline
Pre-conditions	Mobile in airplane mode.		
Screen flow / Steps	**1.** Fill report; submit.		
Expected UI behaviour	OutboxService.enqueue stores SITE_REPORT_CREATE row with photos. Toast: 'Queued — will upload when online.'		
Construction context	—	**Flutter gap**	—

ID	F-SR-004	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Web variant has no offline queue	**Role**	SITE_ENGINEER (web)
Priority	P1	**Type**	Offline Gap
Pre-conditions	Browser; airplane mode (DevTools network=offline).		
Screen flow / Steps	**1.** Submit report.		
Expected UI behaviour	createReportDirect throws; toast: 'Network error — please try again'. No queue.		
Construction context	—	**Flutter gap**	F-G30 — web cannot use OutboxService (drift_flutter unavailable).

ID	F-SR-007	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Cross-project lazy summary on empty list	**Role**	SITE_ENGINEER
Priority	P3	**Type**	UX
Pre-conditions	No reports for selected project; user has submitted to another project today.		
Screen flow / Steps	**1.** Open Site Reports for empty project.		
Expected UI behaviour	Empty card + footer hint 'You submitted 2 reports today to — wrong project?'		
Construction context	—	**Flutter gap**	—

ID	F-SR-008	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Multi-photo upload caps individual size	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Validation
Pre-conditions	Mobile.		
Screen flow / Steps	**1.** Pick 12 photos including one 40 MB raw camera shot.		
Expected UI behaviour	imageQuality=70 compresses; per-file size limit (e.g., 10 MB) enforced before queueing.		
Construction context	—	**Flutter gap**	F-G34 — no explicit file size cap surfaced in UI.

ID	F-SR-009	**Screen**	lib/features/site_reports/...
Title	GPS spoofing detection	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Security Gap
Pre-conditions	User runs mock-location app.		
Screen flow / Steps	**1.** Set device to mock coordinates near project. **2.** Submit report.		
Expected UI behaviour	Today: not detected. Goal: check Position.isMocked (mobile geolocator) and flag for review.		
Construction context	—	**Flutter gap**	F-G39 — no mock-location detection.

ID	F-SR-010	**Screen**	lib/features/site_reports/presentation/screens/site_reports_screen.dart
Title	Photo carousel shows 4 thumbs + '+N' count	**Role**	any
Priority	P3	**Type**	UI
Pre-conditions	Report with 7 photos.		
Screen flow / Steps	**1.** Open list.		
Expected UI behaviour	Card shows 4 thumbnails and '+3' badge.		
Construction context	—	**Flutter gap**	—

ID	F-OFF-002	**Screen**	lib/features/site_reports/...
Title	Storage full mid-photo capture	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	Device storage 99% full.		
Screen flow / Steps	**1.** Try to capture a photo.		
Expected UI behaviour	ImagePicker fails gracefully; UI shows 'Storage full — free up space'.		
Construction context	—	**Flutter gap**	F-G18 — verify error handling on PickerException.

ID	F-PERM-001	**Screen**	lib/features/site_reports/...
Title	Camera permission denied — actionable recovery	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Permission
Pre-conditions	Camera permission previously denied 'Don't ask again'.		
Screen flow / Steps	**1.** Tap 'Capture photo'.		
Expected UI behaviour	Dialog 'Camera disabled — Open Settings'; tap launches OS settings deep link. Returning enables capture.		
Construction context	—	**Flutter gap**	F-G18 — verify permission_handler hook.

ID	F-PERM-002	**Screen**	lib/features/site_reports/...
Title	Storage / Photos permission for gallery picker (Android 13+)	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Permission
Pre-conditions	Android 13 device; READ_MEDIA_IMAGES not granted.		
Screen flow / Steps	**1.** Tap 'Pick from gallery'.		
Expected UI behaviour	Granular media permission prompt (not legacy READ_EXTERNAL_STORAGE). Picker opens after grant.		
Construction context	—	**Flutter gap**	—

ID	F-EVID-001	**Screen**	lib/features/site_reports/...
Title	EXIF stripped from photo before upload (privacy)	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Privacy
Pre-conditions	Photo captured with personal metadata (device model, geotag).		
Screen flow / Steps	**1.** Submit; intercept request body.		
Expected UI behaviour	Today: EXIF preserved (per code review) to support forensic audit. Option needed: strip personal metadata while preserving project geotag (which is captured server-side too). Decision required.		
Construction context	—	**Flutter gap**	F-G40 — EXIF retention strategy needs product decision.

lib/screens/sync

ID	F-SR-005	**Screen**	lib/screens/sync/pending_sync_screen.dart + SyncService
Title	Sync of queued report on reconnect	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Offline
Pre-conditions	2 queued reports, network restored.		
Screen flow / Steps	**1.** Open Pending Sync screen. **2.** Tap 'Sync Now'.		
Expected UI behaviour	Both rows drain to 'done'; toast 'All synced'. PendingSync screen empty.		
Construction context	—	**Flutter gap**	—

ID	F-SR-006	**Screen**	lib/screens/sync/pending_sync_screen.dart
Title	Permanent failure surfaces in Issues tab	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	Photo blob corrupted (5 failed attempts).		
Screen flow / Steps	**1.** Open Issues tab.		
Expected UI behaviour	Row with error message + Retry + Discard buttons.		
Construction context	—	**Flutter gap**	F-G31 — no bulk discard; one-by-one only.

ID	F-SYNC-001	**Screen**	lib/screens/sync/pending_sync_screen.dart
Title	Queued tab shows pending entries with next-retry time	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Offline
Pre-conditions	2 queued, network up.		
Screen flow / Steps	**1.** Open Pending Sync.		
Expected UI behaviour	Both rows visible with attempt count + nextRetryAt timestamp.		
Construction context	—	**Flutter gap**	—

ID	F-SYNC-002	**Screen**	lib/screens/sync/pending_sync_screen.dart
Title	Sync Now button triggers immediate drain	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Offline
Pre-conditions	3 queued; network up.		
Screen flow / Steps	**1.** Tap 'Sync Now'.		
Expected UI behaviour	SyncService.triggerSyncNow runs; rows transition to done; list empties.		
Construction context	—	**Flutter gap**	—

ID	F-SYNC-003	**Screen**	lib/screens/sync/pending_sync_screen.dart
Title	Per-row Retry/Discard on permanent failure	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	1 permanent failure.		
Screen flow / Steps	**1.** Tap Discard.		
Expected UI behaviour	Confirmation dialog; on confirm row removed; photo file deleted from local storage.		
Construction context	—	**Flutter gap**	—

Site report detail

ID	F-SR-011	**Screen**	Site report detail
Title	Approve report fans webhook + push	**Role**	PM
Priority	P2	**Type**	Integration
Pre-conditions	Report SUBMITTED.		
Screen flow / Steps	**1.** Approve.		
Expected UI behaviour	Status APPROVED; ActivityFeedService logs; FCM push to customer (if subscribed).		
Construction context	—	**Flutter gap**	—

lib/services/sync_service.dart

ID	F-SR-012	**Screen**	lib/services/sync_service.dart
Title	Background sync respects exponential backoff	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	3 transient API failures.		
Screen flow / Steps	**1.** Cause server 500 for 3 attempts.		
Expected UI behaviour	Backoff: 2 → 4 → 8 → 16 → 30 min jittered; max 5 attempts before permanent failure.		
Construction context	—	**Flutter gap**	—

ID	F-OFF-001	**Screen**	lib/services/sync_service.dart
Title	App killed mid-photo-upload — resume on relaunch	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Offline
Pre-conditions	2 photos queued; photo 1 mid-upload.		
Screen flow / Steps	**1.** Force-kill app. **2.** Relaunch.		
Expected UI behaviour	OutboxService rehydrates; SyncService re-attempts. No duplicate from photo 1 (idempotency key).		
Construction context	—	**Flutter gap**	—

ID	F-OFF-005	**Screen**	lib/services/sync_service.dart
Title	Sync conflict — same task progress edited offline by 2 devices	**Role**	2 SITE_ENGINEERS
Priority	P2	**Type**	Concurrency Gap
Pre-conditions	Device A offline, Device B offline. Both update progress for same task.		
Screen flow / Steps	**1.** Bring both online; allow sync.		
Expected UI behaviour	Today: last writer wins. Goal: conflict UI in Pending Sync.		
Construction context	—	**Flutter gap**	F-G23 — no conflict-resolution UI.

ID	F-LIFE-001	**Screen**	lib/services/sync_service.dart
Title	App backgrounded during photo upload — upload continues	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	Multi-photo upload in progress.		
Screen flow / Steps	**1.** Press Home; wait 30s; reopen app.		
Expected UI behaviour	Foreground service (Android) or background URLSession (iOS) completes the upload. Sync screen shows progress on resume.		
Construction context	—	**Flutter gap**	F-G18 — confirm WorkManager / BGTaskScheduler integration.

lib/screens/site_visits

ID	F-SV-001	**Screen**	lib/screens/site_visits/site_visit_check_in_dialog.dart
Title	Check-in dialog requires project + GPS + visit type	**Role**	PM / Site Engineer
Priority	P1	**Type**	Validation
Pre-conditions	At project gate.		
Screen flow / Steps	**1.** Tap '+ Check In' on Site Visits. **2.** Skip GPS capture; tap Submit.		
Expected UI behaviour	Validation: 'Capture location' required. Set GPS; submit succeeds; status PENDING → CHECKED_IN.		
Construction context	—	**Flutter gap**	—

ID	F-SV-002	**Screen**	lib/screens/site_visits/site_visits_screen.dart
Title	Active visit banner appears across screens	**Role**	PM
Priority	P2	**Type**	UX
Pre-conditions	Active CHECKED_IN visit.		
Screen flow / Steps	**1.** Navigate to dashboard, leads, projects.		
Expected UI behaviour	Banner sticky at top of Site Visits screen; but doesn't appear on unrelated screens (could be a Provider-driven global banner).		
Construction context	—	**Flutter gap**	F-G17 — visit banner not globally surfaced.

ID	F-SV-003	**Screen**	lib/screens/site_visits/site_visit_check_out_dialog.dart
Title	Check-out distance from project — high value flag	**Role**	PM
Priority	P2	**Type**	Construction-Domain
Pre-conditions	Active visit; check-out from 5km away.		
Screen flow / Steps	**1.** Tap Check Out; capture GPS at far location.		
Expected UI behaviour	Server records distanceFromProjectCheckOut; UI shows '5.0 km from site' chip in red.		
Construction context	—	**Flutter gap**	—

ID	F-SV-004	**Screen**	lib/screens/site_visits/site_visit_check_out_dialog.dart
Title	Visit duration computed client-side and persisted	**Role**	any
Priority	P3	**Type**	Functional
Pre-conditions	Visit checked-in 90 min ago.		
Screen flow / Steps	**1.** Open Check-Out dialog.		
Expected UI behaviour	Header shows '1h 30m'; durationMinutes=90 sent in payload.		
Construction context	—	**Flutter gap**	—

ID	F-SV-005	**Screen**	lib/screens/site_visits/site_visit_detail_screen.dart
Title	Visit detail screen shows full GPS lat/lng	**Role**	any
Priority	P2	**Type**	UI
Pre-conditions	Completed visit.		
Screen flow / Steps	**1.** Open detail.		
Expected UI behaviour	Two timeline cards (check-in / check-out) with 6-decimal coords in monospace.		
Construction context	—	**Flutter gap**	—

ID	F-SV-006	**Screen**	lib/screens/site_visits/...
Title	GPS spoofing for fake on-site visit	**Role**	SITE_ENGINEER
Priority	P1	**Type**	Security Gap
Pre-conditions	Mock location app.		
Screen flow / Steps	**1.** Check in from 100 km away with mocked coords near project.		
Expected UI behaviour	Today: not detected. Goal: flag visit, alert PM.		
Construction context	—	**Flutter gap**	F-G39 — no mock-location check.

lib/features/observations

ID	F-OBS-001	**Screen**	lib/features/observations/presentation/screens/observations_screen.dart
Title	Observations tabs — Active vs Resolved	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	3 active, 2 resolved observations.		
Screen flow / Steps	**1.** Switch tabs.		
Expected UI behaviour	Active tab lists 3; Resolved lists 2. PDF export from app bar.		
Construction context	—	**Flutter gap**	—

ID	F-OBS-002	**Screen**	lib/features/observations/... + task progress screen
Title	Critical observation blocks task completion (cross-screen)	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Integration
Pre-conditions	Critical OPEN observation on Task T.		
Screen flow / Steps	**1.** Try to Mark Complete T.		
Expected UI behaviour	Backend rejects; UI shows 'Critical observation #N must be resolved first'. Action link to observation.		
Construction context	—	**Flutter gap**	F-G18 — confirm cross-screen link works.

lib/features/quality

ID	F-QC-001	**Screen**	lib/features/quality/presentation/screens/quality_checks_screen.dart
Title	Quality check FAIL auto-creates observation	**Role**	QS
Priority	P2	**Type**	Integration
Pre-conditions	QC checklist for slab pour.		
Screen flow / Steps	**1.** Record check with result=FAIL.		
Expected UI behaviour	Backend creates linked observation; UI navigates to it or shows toast 'Observation #N created'.		
Construction context	—	**Flutter gap**	—

lib/features/gallery

ID	F-GAL-001	**Screen**	lib/features/gallery/presentation/screens/gallery_screen.dart
Title	Gallery grouped by date; auth-aware image loading	**Role**	any
Priority	P3	**Type**	Functional
Pre-conditions	Project with 50 gallery images across 5 days.		
Screen flow / Steps	**1.** Open Gallery; toggle grid ↔ list view.		
Expected UI behaviour	Date group headers; images load via authenticated requests (Authorization header). Pull-to-refresh refreshes.		
Construction context	—	**Flutter gap**	—

lib/features/delays

ID	F-DEL-001	**Screen**	lib/features/delays/presentation/screens/delay_logs_screen.dart
Title	Log delay with customer-visible flag	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Project in CONSTRUCTION phase.		
Screen flow / Steps	**1.** Tap 'Log Delay'. **2.** Category=WEATHER, fromDate=today, durationDays=3. **3.** Toggle customerVisible=true; leave customer summary blank. **4.** Submit.		
Expected UI behaviour	Validation: customer summary required when visible flag on. Fill summary; submit. Outbox-queued on mobile.		
Construction context	Monsoon weather delay scenario.	**Flutter gap**	—

ID	F-DEL-002	**Screen**	lib/features/delays/presentation/screens/delay_logs_screen.dart
Title	Close delay with toDate after fromDate	**Role**	PM
Priority	P2	**Type**	Validation
Pre-conditions	Open delay starting 5 days ago.		
Screen flow / Steps	**1.** Tap Close; pick toDate=yesterday.		
Expected UI behaviour	Date picker prevents toDate < fromDate. Submit succeeds; delay row shows duration.		
Construction context	—	**Flutter gap**	—

lib/screens/labour

ID	F-LAB-001	**Screen**	lib/screens/labour/attendance_screen.dart
Title	Bulk attendance — 50 labourers in 30 s	**Role**	SITE_SUPERVISOR
Priority	P1	**Type**	Functional
Pre-conditions	50 labour records active on project.		
Screen flow / Steps	**1.** Select project + today. **2.** Mark 45 PRESENT, 3 HALF_DAY, 2 ABSENT via chips. **3.** Submit.		
Expected UI behaviour	50 attendance rows persisted; success toast.		
Construction context	—	**Flutter gap**	F-G19 — no RFID/Aadhaar capture; chip taps only.

ID	F-LAB-002	**Screen**	lib/screens/labour/attendance_screen.dart
Title	Duplicate attendance entry handled	**Role**	SITE_SUPERVISOR
Priority	P2	**Type**	Negative
Pre-conditions	Attendance for today already exists.		
Screen flow / Steps	**1.** Try to re-submit.		
Expected UI behaviour	Backend 409; UI shows 'Already recorded today'.		
Construction context	—	**Flutter gap**	—

ID	F-LAB-003	**Screen**	lib/screens/labour/add_labour_screen.dart
Title	Add Labour — wage decimal + ID proof type	**Role**	HR / SUPERVISOR
Priority	P2	**Type**	Validation
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Fill name, phone, trade=MASON, dailyWage=850.50, idProofType=AADHAAR, idProofNumber='1234 5678 9012'. **2.** Save.		
Expected UI behaviour	Stored; Aadhaar masked in UI to '**** **** 9012'.		
Construction context	—	**Flutter gap**	F-G29 — no client-side Aadhaar verification/masking before upload (relies on backend).

ID	F-LAB-004	**Screen**	lib/screens/labour/mb_entry_screen.dart
Title	MB entry auto-calculates qty from L×B×D	**Role**	SUPERVISOR
Priority	P2	**Type**	Functional
Pre-conditions	MB form open.		
Screen flow / Steps	**1.** Enter L=4, B=2, D=0 (footing). **2.** Observe Quantity field.		
Expected UI behaviour	Qty = 4 × 2 × max(0,1) = 8 SFT. (Code multiplies 0 → 1 to avoid zeroing.)		
Construction context	Foundation footing measurement.	**Flutter gap**	—

ID	F-LAB-005	**Screen**	lib/screens/labour/mb_list_screen.dart
Title	MB list filters by project	**Role**	SUPERVISOR
Priority	P3	**Type**	Functional
Pre-conditions	MB entries across 2 projects.		
Screen flow / Steps	**1.** Change project dropdown.		
Expected UI behaviour	List reloads; totals update.		
Construction context	—	**Flutter gap**	—

ID	F-IND-005	**Screen**	lib/screens/labour/labour_list_screen.dart
Title	Aadhaar masked in labour list	**Role**	any
Priority	P1	**Type**	Privacy
Pre-conditions	Labour with idProofType=AADHAAR.		
Screen flow / Steps	**1.** Open list / detail.		
Expected UI behaviour	Aadhaar shown as '**** **** 9012' (last 4 only).		
Construction context	—	**Flutter gap**	F-G29 — verify masking in labour detail.

lib/features/labour

ID	F-LAB-006	**Screen**	lib/features/labour/presentation/screens/wage_sheet_screen.dart
Title	Wage sheet generation respects date range	**Role**	HR / Finance
Priority	P1	**Type**	Functional
Pre-conditions	Attendance for 26 working days.		
Screen flow / Steps	**1.** Pick start/end dates; tap Generate.		
Expected UI behaviour	WageSheet DRAFT; one row per labourer; daysWorked, wage, advances, net payable.		
Construction context	—	**Flutter gap**	F-G18 — no PF/ESI deductions visible (gap from backend).

lib/screens/inventory

ID	F-INV-001	**Screen**	lib/screens/inventory/materials_screen.dart
Title	Materials list — add with category + unit	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Materials screen open.		
Screen flow / Steps	**1.** Add 'TMT Steel Fe500', category=METAL, unit=KG.		
Expected UI behaviour	Persists; new row visible.		
Construction context	—	**Flutter gap**	F-G08 — no HSN field on material form.

ID	F-INV-002	**Screen**	lib/screens/inventory/add_stock_adjustment_screen.dart
Title	Stock adjustment requires reason	**Role**	STORE_KEEPER
Priority	P2	**Type**	Validation
Pre-conditions	Stock for Cement = 100 bags.		
Screen flow / Steps	**1.** Adjust -10 with reason blank; Save.		
Expected UI behaviour	Form validation: reason required.		
Construction context	—	**Flutter gap**	F-G18 — no approver workflow gate.

ID	F-INV-003	**Screen**	lib/screens/inventory/material_consumption_screen.dart
Title	Material consumption report rows	**Role**	STORE_KEEPER
Priority	P3	**Type**	Functional
Pre-conditions	Material with inward 200, current 50, wastage 5.		
Screen flow / Steps	**1.** Open consumption report.		
Expected UI behaviour	Row: 200 / 50 / 5 across columns.		
Construction context	—	**Flutter gap**	—

ID	F-INV-004	**Screen**	lib/screens/inventory/stock_report_screen.dart
Title	Stock report empty state	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	No data.		
Screen flow / Steps	**1.** Open report.		
Expected UI behaviour	Centered icon + 'No stock data' message.		
Construction context	—	**Flutter gap**	—

ID	F-IND-004	**Screen**	lib/screens/inventory/add_material_screen.dart
Title	HSN/SAC code input on material	**Role**	ADMIN
Priority	P1	**Type**	India Compliance Gap
Pre-conditions	Add material.		
Screen flow / Steps	**1.** Fill name/unit/rate; look for HSN/SAC.		
Expected UI behaviour	HSN/SAC required input (4-8 digits per backend G-21).		
Construction context	—	**Flutter gap**	F-G08 — HSN/SAC field missing on material form.

lib/screens/procurement

ID	F-PRC-001	**Screen**	lib/screens/procurement/add_vendor_screen.dart
Title	Add vendor — GSTIN length	**Role**	PROCUREMENT
Priority	P1	**Type**	Validation / India
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Type GSTIN '29ABC' (5 chars); Save.		
Expected UI behaviour	Validator: 'GSTIN must be 15 chars'.		
Construction context	—	**Flutter gap**	F-G06 — no checksum / live GSTN validation.

ID	F-PRC-002	**Screen**	lib/screens/procurement/po_list_screen.dart
Title	PO list status filter + pagination	**Role**	PROCUREMENT
Priority	P2	**Type**	Functional
Pre-conditions	POs in DRAFT/ISSUED/RECEIVED.		
Screen flow / Steps	**1.** Filter='ISSUED'.		
Expected UI behaviour	Only issued POs; pagination updates.		
Construction context	—	**Flutter gap**	—

ID	F-PRC-003	**Screen**	lib/screens/procurement/add_purchase_order_screen.dart
Title	Add PO — Indian numbering on total	**Role**	PROCUREMENT
Priority	P3	**Type**	Localisation
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Enter line items totalling ■12,34,56,789.		
Expected UI behaviour	Total formatted as Indian grouping (lakh/crore). Today: raw number.		
Construction context	—	**Flutter gap**	F-G28 — no Indian numbering helper.

ID	F-PRC-005	**Screen**	lib/screens/procurement/add_vendor_screen.dart
Title	Vendor compliance — phone uniqueness	**Role**	PROCUREMENT
Priority	P2	**Type**	Negative
Pre-conditions	Vendor V1 has phone 9999.		
Screen flow / Steps	**1.** Create V2 with same phone.		
Expected UI behaviour	Backend 409; UI surfaces 'Vendor with this phone exists'.		
Construction context	—	**Flutter gap**	—

lib/features/procurement

ID	F-PRC-004	**Screen**	lib/features/procurement/presentation/screens/quotation_management_screen.dart
Title	Quotation management — select winner; others auto-reject	**Role**	PROCUREMENT
Priority	P1	**Type**	Functional
Pre-conditions	3 quotations on one indent.		
Screen flow / Steps	**1.** Tap 'Select' on Q2.		
Expected UI behaviour	Q2 SELECTED; Q1, Q3 REJECTED; selectedAt stored.		
Construction context	—	**Flutter gap**	—

ID	F-PROC-RFQ-001	**Screen**	lib/features/procurement/presentation/screens/quotation_management_screen.dart
Title	Vendor quotation comparison side-by-side	**Role**	PROCUREMENT
Priority	P2	**Type**	Functional
Pre-conditions	3 quotations on indent.		
Screen flow / Steps	**1.** Open quotation comparison.		
Expected UI behaviour	Table with 3 columns: amount, delivery, taxes, validity. Highlight lowest amount. Select winner button.		
Construction context	—	**Flutter gap**	—

lib/features/boq

ID	F-BOQ-001	**Screen**	lib/features/boq/presentation/screens/boq_screen.dart
Title	BOQ list — currency formatting & summary card	**Role**	PM / Estimator
Priority	P2	**Type**	Functional
Pre-conditions	BOQ items present.		
Screen flow / Steps	**1.** Open BOQ.		
Expected UI behaviour	Items grouped by category; totals shown as ■ with 2 decimals; summary card toggleable.		
Construction context	—	**Flutter gap**	F-G37 — no GST split / HSN visible per row.

ID	F-BOQ-002	**Screen**	lib/features/boq/presentation/screens/boq_document_management_screen.dart
Title	BOQ doc — submit → approve → state transitions	**Role**	PM (BOQ_APPROVE)
Priority	P1	**Type**	Functional
Pre-conditions	BoQ DRAFT with items.		
Screen flow / Steps	**1.** Tap Submit; observe status PENDING_APPROVAL. **2.** Tap Internal Approve.		
Expected UI behaviour	Status APPROVED; timestamps row updates; further item edits blocked. No confirmation dialog today (gap).		
Construction context	—	**Flutter gap**	F-G21 — no confirm dialog on irreversible approve.

ID	F-BOQ-003	**Screen**	lib/features/boq/presentation/screens/boq_document_management_screen.dart
Title	BOQ reject requires reason	**Role**	PM
Priority	P2	**Type**	Validation
Pre-conditions	BoQ in PENDING_APPROVAL.		
Screen flow / Steps	**1.** Tap Reject; submit with blank reason.		
Expected UI behaviour	Dialog validation refuses; rejection reason required.		
Construction context	—	**Flutter gap**	—

ID	F-BOQ-004	**Screen**	lib/features/boq/presentation/screens/boq_invoice_screen.dart
Title	BOQ invoice — confirm payment requires reference	**Role**	FINANCE
Priority	P1	**Type**	Validation
Pre-conditions	Invoice SENT.		
Screen flow / Steps	**1.** Tap 'Confirm Payment Received'; leave ref blank.		
Expected UI behaviour	Dialog: payment reference required.		
Construction context	—	**Flutter gap**	—

ID	F-BOQ-005	**Screen**	lib/features/boq/presentation/screens/co_management_screen.dart
Title	Change order — addition triggers advance invoice	**Role**	PM (VO_APPROVE)
Priority	P1	**Type**	Functional
Pre-conditions	CO addition ■2,50,000 approved by customer.		
Screen flow / Steps	**1.** Tap 'Start Work'.		
Expected UI behaviour	Status IN_PROGRESS; advance invoice auto-generated; visible in Invoices screen.		
Construction context	—	**Flutter gap**	—

ID	F-BOQ-006	**Screen**	lib/features/boq/presentation/screens/payment_schedule_screen.dart
Title	Payment schedule — raise invoice on DUE stage	**Role**	FINANCE
Priority	P1	**Type**	Functional
Pre-conditions	Stage 2 in DUE.		
Screen flow / Steps	**1.** Tap 'Raise Invoice'; pick today as invoice date.		
Expected UI behaviour	Tax invoice created with stored GST rate; status moves to INVOICED.		
Construction context	—	**Flutter gap**	F-G22 — no idempotency token on tap (network retry could duplicate).

ID	F-BOQ-007	**Screen**	lib/features/boq/presentation/screens/payment_schedule_screen.dart
Title	Finance summary card reconciles after CO	**Role**	FINANCE
Priority	P2	**Type**	Integration
Pre-conditions	Project with 1 addition CO ■1.2L approved.		
Screen flow / Steps	**1.** Open Payment Schedule.		
Expected UI behaviour	Net project value = original + addition; collected/outstanding computed from invoices + receipts.		
Construction context	—	**Flutter gap**	—

ID	F-BOQ-008	**Screen**	lib/features/boq/...
Title	CGST/SGST/IGST split UI missing	**Role**	FINANCE
Priority	P1	**Type**	India Compliance Gap
Pre-conditions	Intra- vs inter-state invoice.		
Screen flow / Steps	**1.** Open invoice screen for KA→KA project.		
Expected UI behaviour	Should show CGST 9% + SGST 9% split. Today: only single 'GST 18%' line. Filing GSTR-1 with this data impossible without backend splitting.		
Construction context	—	**Flutter gap**	F-G06 — no CGST/SGST/IGST split anywhere in UI.

ID	F-IND-001	**Screen**	lib/features/boq/.../payment_schedule_screen.dart
Title	Indian number formatting in financial dashboard	**Role**	FINANCE
Priority	P3	**Type**	Localisation
Pre-conditions	Total = ■12,34,56,789.		
Screen flow / Steps	**1.** Open payment schedule.		
Expected UI behaviour	Displayed '12,34,56,789' (Indian grouping). Today: many screens raw or US grouping.		
Construction context	—	**Flutter gap**	F-G28 — no shared lakh/crore formatter.

ID	F-IND-002	**Screen**	lib/features/boq/.../boq_invoice_screen.dart
Title	GST split UI on invoice — gap	**Role**	FINANCE
Priority	P1	**Type**	India Compliance Gap
Pre-conditions	Intra-state invoice.		
Screen flow / Steps	**1.** Open invoice detail.		
Expected UI behaviour	CGST 9% + SGST 9% lines. Today: single GST line.		
Construction context	—	**Flutter gap**	F-G06 — CGST/SGST/IGST split absent.

ID	F-LONG-001	**Screen**	lib/features/boq/presentation/screens/boq_screen.dart
Title	Specifications field 5000-char limit on BOQ	**Role**	ESTIMATOR
Priority	P3	**Type**	Validation
Pre-conditions	Create BOQ item.		
Screen flow / Steps	**1.** Paste 6000-char text into specifications.		
Expected UI behaviour	Server validation rejects; UI should show counter '5000/5000'.		
Construction context	—	**Flutter gap**	F-G18 — counter widget not visible on screen.

lib/features/dpc

ID	F-DPC-001	**Screen**	lib/features/dpc/presentation/screens/dpc_builder_screen.dart
Title	DPC builder requires approved BOQ	**Role**	PM / Estimator
Priority	P1	**Type**	Functional
Pre-conditions	Project with DRAFT BOQ.		
Screen flow / Steps	**1.** Open DPC builder.		
Expected UI behaviour	Empty state 'No approved BoQ — approve the BOQ first' with link to BOQ doc screen.		
Construction context	—	**Flutter gap**	—

ID	F-DPC-002	**Screen**	lib/features/dpc/presentation/screens/dpc_builder_screen.dart
Title	Split-screen PDF preview on wide screens	**Role**	any
Priority	P2	**Type**	Responsive
Pre-conditions	Approved BOQ; window > 1200px.		
Screen flow / Steps	**1.** Edit a scope rationale.		
Expected UI behaviour	Right pane re-renders PDF preview after debounce. Below 1200px preview stacks below content.		
Construction context	—	**Flutter gap**	—

ID	F-DPC-003	**Screen**	lib/features/dpc/presentation/screens/dpc_builder_screen.dart
Title	Issue DPC freezes document	**Role**	ADMIN
Priority	P1	**Type**	Functional
Pre-conditions	DRAFT DPC.		
Screen flow / Steps	**1.** Edit content; tap 'Issue'.		
Expected UI behaviour	Status DRAFT → ISSUED; PDF persisted; further edits blocked until 'Revise'.		
Construction context	—	**Flutter gap**	F-G21 — no confirm dialog before irreversible Issue.

ID	F-DPC-004	**Screen**	lib/features/dpc/presentation/screens/dpc_revisions_screen.dart
Title	Revisions screen — download prior PDF	**Role**	any
Priority	P2	**Type**	Functional
Pre-conditions	2 revisions issued.		
Screen flow / Steps	**1.** Tap 'Download' on revision 1.		
Expected UI behaviour	PDF fetched and saved/opened.		
Construction context	—	**Flutter gap**	F-G24 — no in-app PDF preview before download.

ID	F-DPC-006	**Screen**	lib/features/dpc/presentation/screens/dpc_builder_screen.dart
Title	Catalog rate override on DPC line — audit gap	**Role**	ADMIN
Priority	P2	**Type**	Compliance Gap
Pre-conditions	Catalog item rate ■1200.		
Screen flow / Steps	**1.** Add customisation line; override unit rate to ■600. **2.** Save.		
Expected UI behaviour	Today: accepted silently. Goal: audit log entry + reviewer approval required.		
Construction context	—	**Flutter gap**	F-G18 — no override audit; ties to backend G-09.

ID	F-RSP-002	**Screen**	lib/features/dpc/presentation/screens/dpc_builder_screen.dart
Title	DPC builder stacks below 1200px	**Role**	any
Priority	P3	**Type**	Responsive
Pre-conditions	Builder open at 1199px.		
Screen flow / Steps	**1.** Increase width past 1200px.		
Expected UI behaviour	PDF preview pane attaches to the right.		
Construction context	—	**Flutter gap**	—

lib/features/dpc_customization_catalog

ID	F-DPC-005	**Screen**	lib/features/dpc_customization_catalog/presentation/screens/dpc_customization_catalog_admin_screen.dart
Title	Customization catalog admin — search debounce	**Role**	ADMIN
Priority	P3	**Type**	Performance
Pre-conditions	500 catalog items.		
Screen flow / Steps	**1.** Type 'kitchen' rapidly.		
Expected UI behaviour	Debounced search; page resets to 0; results filter.		
Construction context	—	**Flutter gap**	—

lib/features/estimation_settings

ID	F-ESET-001	**Screen**	lib/features/estimation_settings/presentation/screens/packages_list_screen.dart
Title	Packages list — show/hide inactive	**Role**	ADMIN
Priority	P3	**Type**	Functional
Pre-conditions	Mixed active/inactive packages.		
Screen flow / Steps	**1.** Toggle 'Show inactive'.		
Expected UI behaviour	List re-fetches; inactive shown.		
Construction context	—	**Flutter gap**	—

ID	F-ESET-002	**Screen**	lib/features/estimation_settings/presentation/screens/market_index_screen.dart
Title	Market index — publish new snapshot	**Role**	ADMIN (canPublishMarketIndex)
Priority	P2	**Type**	Functional
Pre-conditions	Last published 6 months ago.		
Screen flow / Steps	**1.** Tap 'New Snapshot'. **2.** Fill rates for steel, cement, etc. **3.** Submit.		
Expected UI behaviour	Composite index computed; row inserted; Active chip on it.		
Construction context	—	**Flutter gap**	—

ID	F-ESET-003	**Screen**	lib/features/estimation_settings/presentation/screens/rate_card_screen.dart
Title	Rate card version creation	**Role**	ADMIN
Priority	P3	**Type**	Functional
Pre-conditions	Package + project type selected.		
Screen flow / Steps	**1.** Tap 'New Version'. **2.** Submit material/labour/overhead rates with valid dates.		
Expected UI behaviour	Version appears with Active chip if effective-today.		
Construction context	—	**Flutter gap**	—

lib/screens/finance

ID	F-FIN-001	**Screen**	lib/screens/finance/finance_dashboard_screen.dart
Title	Finance dashboard tabs	**Role**	FINANCE
Priority	P2	**Type**	Functional
Pre-conditions	Mixed invoices, bills, labour payments.		
Screen flow / Steps	**1.** Switch tabs.		
Expected UI behaviour	Each tab loads independently; counts visible.		
Construction context	—	**Flutter gap**	F-G37 — no summary totals; no status filtering visible.

ID	F-FIN-002	**Screen**	lib/screens/finance/add_project_invoice_screen.dart
Title	Add project invoice — line items with GST	**Role**	FINANCE
Priority	P1	**Type**	India Compliance
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Add lines with taxable values; choose GST 18%.		
Expected UI behaviour	Total + GST + grand total computed. Should split CGST+SGST or IGST based on customer state — today not done.		
Construction context	—	**Flutter gap**	F-G06 — no CGST/SGST/IGST split.

lib/screens/payments

ID	F-PAY-001	**Screen**	lib/screens/payments/payments_dashboard_screen.dart
Title	Payments dashboard — pending vs all chips	**Role**	FINANCE
Priority	P2	**Type**	Functional
Pre-conditions	Mixed pending + paid.		
Screen flow / Steps	**1.** Tap 'Pending'.		
Expected UI behaviour	List filters to non-PAID; pagination resets.		
Construction context	—	**Flutter gap**	—

ID	F-PAY-002	**Screen**	lib/screens/payments/challan_management_screen.dart
Title	Challan management — auto numbering per FY	**Role**	FINANCE
Priority	P1	**Type**	India Compliance
Pre-conditions	3 challans created today (FY 2026-27).		
Screen flow / Steps	**1.** Create one more.		
Expected UI behaviour	Number is WAL/CH/2026-27/00004; backend pessimistic lock.		
Construction context	—	**Flutter gap**	—

ID	F-PAY-003	**Screen**	lib/screens/payments/payment_history_screen.dart
Title	Payment history — date range filter	**Role**	FINANCE
Priority	P2	**Type**	Functional
Pre-conditions	Year of receipts.		
Screen flow / Steps	**1.** Pick Apr 1–Sep 30.		
Expected UI behaviour	Receipts filtered; total at footer.		
Construction context	—	**Flutter gap**	—

lib/features/stage_payments

ID	F-STG-001	**Screen**	lib/features/stage_payments/presentation/screens/stage_payment_screen.dart
Title	Stage timeline status pills	**Role**	PM / FINANCE
Priority	P2	**Type**	UI
Pre-conditions	Stages UPCOMING/DUE/INVOICED/PAID/OVERDUE.		
Screen flow / Steps	**1.** Open timeline.		
Expected UI behaviour	Pills colour-coded grey/orange/blue/green/red.		
Construction context	—	**Flutter gap**	—

ID	F-STG-002	**Screen**	lib/features/stage_payments/presentation/screens/stage_payment_screen.dart
Title	Certify stage requires 'certified by'	**Role**	PM / DIRECTOR
Priority	P1	**Type**	Validation
Pre-conditions	Stage UPCOMING.		
Screen flow / Steps	**1.** Tap Certify; leave certified-by blank.		
Expected UI behaviour	Dialog validation requires field. Provide name; default retention 5%; submit.		
Construction context	—	**Flutter gap**	F-G21 — no confirmation dialog before final certification.

ID	F-STG-003	**Screen**	lib/features/stage_payments/presentation/screens/stage_payment_screen.dart
Title	Certify stage idempotency on rapid double-tap	**Role**	PM
Priority	P1	**Type**	Race
Pre-conditions	Stage DUE.		
Screen flow / Steps	**1.** Tap Certify twice rapidly.		
Expected UI behaviour	Today: backend may create duplicate event. Goal: button disable + idempotency key.		
Construction context	—	**Flutter gap**	F-G22 — no idempotency token; ties to backend G-18.

lib/features/deductions

ID	F-DED-001	**Screen**	lib/features/deductions/presentation/screens/deduction_register_screen.dart
Title	Deduction PARTIALLY_ACCEPTABLE requires amount <= requested	**Role**	DIRECTOR
Priority	P1	**Type**	Validation
Pre-conditions	Deduction requested 60k.		
Screen flow / Steps	**1.** Decide PARTIAL with acceptedAmount=70k.		
Expected UI behaviour	UI rejects; show error message.		
Construction context	—	**Flutter gap**	—

ID	F-DED-002	**Screen**	lib/features/deductions/presentation/screens/deduction_register_screen.dart
Title	Decision dialog requires 'approved by'	**Role**	DIRECTOR
Priority	P2	**Type**	Validation
Pre-conditions	Pending deduction.		
Screen flow / Steps	**1.** Decide ACCEPT; submit without approver name.		
Expected UI behaviour	Dialog validation blocks.		
Construction context	—	**Flutter gap**	—

lib/features/final_account

ID	F-FA-001	**Screen**	lib/features/final_account/presentation/screens/final_account_screen.dart
Title	Final account create requires base + preparedBy	**Role**	COMMERCIAL_MANAGER
Priority	P1	**Type**	Validation
Pre-conditions	No FA yet.		
Screen flow / Steps	**1.** Tap 'Create'. **2.** Leave base contract value blank; submit.		
Expected UI behaviour	Validation refuses. Fill values + preparedBy; submit; status DRAFT.		
Construction context	—	**Flutter gap**	—

ID	F-FA-002	**Screen**	lib/features/final_account/presentation/screens/final_account_screen.dart
Title	Transition to AGREED requires agreedBy	**Role**	DIRECTOR
Priority	P1	**Type**	Validation
Pre-conditions	FA SUBMITTED.		
Screen flow / Steps	**1.** Tap 'Agree'; submit blank.		
Expected UI behaviour	Dialog requires agreedBy. Submit succeeds; status AGREED; DLP starts.		
Construction context	—	**Flutter gap**	F-G21 — no confirmation that AGREED is essentially irreversible.

ID	F-FA-003	**Screen**	lib/features/final_account/presentation/screens/final_account_screen.dart
Title	Optimistic-lock conflict surfaces 409 in UI	**Role**	FINANCE
Priority	P2	**Type**	Concurrency
Pre-conditions	Two users editing same FA.		
Screen flow / Steps	**1.** Both edit + save.		
Expected UI behaviour	Second save receives 409 (backend G-19 fix); UI shows 'This final account was modified by another user. Please refresh and reapply your changes.'		
Construction context	—	**Flutter gap**	F-G23 — UI must consume the 409 and offer refetch.

lib/features/variation_orders

ID	F-VO-001	**Screen**	lib/features/variation_orders/presentation/screens/vo_list_screen.dart
Title	VO list — filter by status; ± amount colour	**Role**	PM
Priority	P2	**Type**	UI
Pre-conditions	Mixed COs.		
Screen flow / Steps	**1.** Filter='APPROVED'.		
Expected UI behaviour	Only approved; addition amounts green, reductions orange.		
Construction context	—	**Flutter gap**	—

ID	F-VO-002	**Screen**	lib/features/variation_orders/presentation/screens/vo_create_screen.dart
Title	VO create — required fields and GST not auto	**Role**	PM
Priority	P2	**Type**	Validation / Gap
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Fill title; submit.		
Expected UI behaviour	Required: title, type, category, net amount, description, justification. GST is NOT auto-calculated from net amount.		
Construction context	—	**Flutter gap**	F-G18 — no GST auto-calc; user must compute mentally.

ID	F-VO-003	**Screen**	lib/features/variation_orders/presentation/screens/vo_detail_screen.dart
Title	VO detail — approve/reject tabs flow	**Role**	DIRECTOR
Priority	P1	**Type**	Functional
Pre-conditions	VO SUBMITTED.		
Screen flow / Steps	**1.** Tap Approve; comment 'OK to proceed'.		
Expected UI behaviour	Backend transitions to INTERNALLY_APPROVED; UI refreshes.		
Construction context	—	**Flutter gap**	—

ID	F-VO-004	**Screen**	lib/features/variation_orders/presentation/screens/vo_detail_screen.dart
Title	VO PDF generation	**Role**	any
Priority	P3	**Type**	Gap
Pre-conditions	VO in any status.		
Screen flow / Steps	**1.** Look for PDF / Share buttons.		
Expected UI behaviour	Today: no PDF generation visible. Customer must print screenshot.		
Construction context	—	**Flutter gap**	F-G24 — VO PDF export missing.

lib/screens/notifications

ID	F-NOTI-001	**Screen**	lib/screens/notifications/portal_notification_screen.dart
Title	Mark all read clears badge	**Role**	any
Priority	P2	**Type**	Functional
Pre-conditions	10 unread.		
Screen flow / Steps	**1.** Open notifications. **2.** Tap 'Mark all read'.		
Expected UI behaviour	Badge clears; PUT /read-all called; list opacity changes.		
Construction context	—	**Flutter gap**	—

ID	F-NOTI-002	**Screen**	lib/screens/notifications/portal_notification_screen.dart
Title	Pull-to-refresh and infinite scroll	**Role**	any
Priority	P3	**Type**	Functional
Pre-conditions	100 notifications.		
Screen flow / Steps	**1.** Pull down; scroll to bottom.		
Expected UI behaviour	Refresh fetches latest; _hasMore loads next page.		
Construction context	—	**Flutter gap**	F-G31 — no type filter; no delete/archive.

lib/screens/support

ID	F-SUP-001	**Screen**	lib/screens/support/support_tickets_screen.dart
Title	Ticket list status + category filters	**Role**	SUPPORT_AGENT
Priority	P2	**Type**	Functional
Pre-conditions	Mixed tickets.		
Screen flow / Steps	**1.** Filter='OPEN', category='BILLING'.		
Expected UI behaviour	List filtered; pagination updates.		
Construction context	—	**Flutter gap**	F-G32 — no SLA badge in list.

ID	F-SUP-002	**Screen**	lib/screens/support/support_ticket_detail_screen.dart
Title	Ticket detail — assignee dropdown + reply	**Role**	SUPPORT_AGENT
Priority	P2	**Type**	Functional
Pre-conditions	Ticket OPEN.		
Screen flow / Steps	**1.** Pick assignee. **2.** Type reply; Send.		
Expected UI behaviour	Reply appears in thread; assignee updated.		
Construction context	—	**Flutter gap**	F-G32 — no attachment in reply UI.

lib/features/feedback

ID	F-FB-001	**Screen**	lib/features/feedback/presentation/screens/feedback_screen.dart
Title	Feedback forms tab vs responses	**Role**	PM
Priority	P3	**Type**	Functional
Pre-conditions	Project with form + responses.		
Screen flow / Steps	**1.** Switch tabs.		
Expected UI behaviour	Forms tab shows form summaries; Responses tab shows submitted data.		
Construction context	—	**Flutter gap**	—

lib/features/partnerships

ID	F-PRT-001	**Screen**	lib/features/partnerships/presentation/screens/partnerships_admin_screen.dart
Title	Partner admin list — tab + type filter + search	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Partners across statuses.		
Screen flow / Steps	**1.** Tab='pending'; type='architect'; search='Anand'.		
Expected UI behaviour	Composed filter list.		
Construction context	—	**Flutter gap**	—

ID	F-PRT-002	**Screen**	lib/features/partnerships/presentation/screens/partner_admin_detail_screen.dart
Title	Partner detail — approve workflow	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Partner pending.		
Screen flow / Steps	**1.** Tap Approve; confirm.		
Expected UI behaviour	Status active; tab badge counts update.		
Construction context	—	**Flutter gap**	F-G18 — no document verify / KYC checklist UI.

lib/features/warranties

ID	F-WAR-001	**Screen**	lib/features/warranties/presentation/screens/warranties_screen.dart
Title	Warranty list shows DLP timer	**Role**	PM / Customer
Priority	P2	**Type**	Construction-Domain Gap
Pre-conditions	Warranty start = handover; period 12 months.		
Screen flow / Steps	**1.** Open warranties.		
Expected UI behaviour	Each row shows 'Expires in N days' chip; alert if < 30 days.		
Construction context	—	**Flutter gap**	F-G33 — DLP countdown not visible; no expiry alert.

ID	F-WAR-002	**Screen**	lib/features/warranties/presentation/screens/warranties_screen.dart
Title	Add warranty (PM)	**Role**	PM
Priority	P3	**Type**	Functional
Pre-conditions	Project COMPLETED.		
Screen flow / Steps	**1.** Tap '+ Warranty'. **2.** Pick component, vendor, period. **3.** Save.		
Expected UI behaviour	Row inserted.		
Construction context	—	**Flutter gap**	—

lib/screens/documents

ID	F-DOC-001	**Screen**	lib/screens/documents/project_document_list_screen.dart
Title	Project document upload + categorisation	**Role**	PM
Priority	P2	**Type**	Functional
Pre-conditions	Project documents tab.		
Screen flow / Steps	**1.** Tap upload FAB; pick file; choose category; submit.		
Expected UI behaviour	Document appears in list; category chip applied.		
Construction context	—	**Flutter gap**	F-G34 — no upload progress indicator.

ID	F-DOC-002	**Screen**	lib/screens/documents/approval_center_screen.dart
Title	Approval centre — filter + pagination	**Role**	DIRECTOR
Priority	P2	**Type**	Functional
Pre-conditions	Approvals across statuses.		
Screen flow / Steps	**1.** Filter='PENDING'.		
Expected UI behaviour	List filters; detail navigation today shows snackbar stub.		
Construction context	—	**Flutter gap**	F-G11 — approval detail screen not implemented.

ID	F-DOC-004	**Screen**	lib/screens/documents/design_agreement_screen.dart
Title	Create document — design agreement form	**Role**	ADMIN / PM
Priority	P3	**Type**	Functional
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Fill client name, area, fee/sqft.		
Expected UI behaviour	Form validates GlobalKey; save generates document.		
Construction context	—	**Flutter gap**	—

ID	F-SUB-RFQ-001	**Screen**	lib/screens/documents/approval_center_screen.dart
Title	Approval centre — bulk approve with same reason	**Role**	DIRECTOR
Priority	P2	**Type**	UX gap
Pre-conditions	6 similar approvals pending.		
Screen flow / Steps	**1.** Select multiple rows; tap Approve.		
Expected UI behaviour	Today: no bulk action. Goal: checkbox + bulk approve dialog.		
Construction context	—	**Flutter gap**	F-G18 — bulk operations missing.

lib/features/shared

ID	F-DOC-003	**Screen**	lib/features/shared/universal_file_viewer_screen.dart
Title	Universal file viewer — PDF + image with auth	**Role**	any
Priority	P2	**Type**	Functional
Pre-conditions	Signed URL.		
Screen flow / Steps	**1.** Open PDF; open JPG.		
Expected UI behaviour	SfPdfViewer renders; image in InteractiveViewer. Authorization header attached via StorageService.		
Construction context	—	**Flutter gap**	F-G24 — no Office doc (.docx/.xlsx) preview; no video.

lib/screens/acl

ID	F-ACL-001	**Screen**	lib/screens/acl/acl_screen.dart
Title	ADMIN role immutable	**Role**	ADMIN
Priority	P1	**Type**	Security
Pre-conditions	ACL screen.		
Screen flow / Steps	**1.** Select ADMIN role.		
Expected UI behaviour	All permission toggles disabled; Save bar hidden; explanation banner shown.		
Construction context	—	**Flutter gap**	—

ID	F-ACL-002	**Screen**	lib/screens/acl/acl_screen.dart
Title	Permission edit — discard vs save	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	PM role selected.		
Screen flow / Steps	**1.** Toggle 2 permissions; tap Discard. **2.** Toggle again; tap Save.		
Expected UI behaviour	Discard reverts; Save persists; 'unsaved changes' badge disappears after Save.		
Construction context	—	**Flutter gap**	—

ID	F-ACL-003	**Screen**	lib/screens/acl/acl_screen.dart
Title	Apply role template confirms	**Role**	ADMIN
Priority	P2	**Type**	Functional
Pre-conditions	Custom role open.		
Screen flow / Steps	**1.** Tap a template chip.		
Expected UI behaviour	Confirmation dialog: 'Replace current permissions with ?'. Confirm applies preset.		
Construction context	—	**Flutter gap**	—

ID	F-ACL-004	**Screen**	lib/screens/acl/acl_screen.dart
Title	Access denied for non-admin	**Role**	SALES
Priority	P1	**Type**	RBAC
Pre-conditions	SALES user.		
Screen flow / Steps	**1.** Try to open /acl.		
Expected UI behaviour	_buildAccessDenied() renders; menu item also hidden.		
Construction context	—	**Flutter gap**	—

lib/screens/portal_users

ID	F-PU-001	**Screen**	lib/screens/portal_users/add_portal_user_screen.dart
Title	Add portal user — role + enabled toggle	**Role**	ADMIN
Priority	P1	**Type**	Functional
Pre-conditions	Add screen.		
Screen flow / Steps	**1.** Fill name/email/password; pick role 'PROJECT_MANAGER'; Save.		
Expected UI behaviour	User created; appears in list.		
Construction context	—	**Flutter gap**	F-G35 — role picker not project-scoped.

ID	F-PU-002	**Screen**	lib/screens/portal_users/portal_users_screen.dart
Title	Portal users list search resets pagination	**Role**	ADMIN
Priority	P3	**Type**	Functional
Pre-conditions	Many users; page=3.		
Screen flow / Steps	**1.** Type 'rao' in search.		
Expected UI behaviour	page resets to 0; matching users shown.		
Construction context	—	**Flutter gap**	—

lib/screens/team_members

ID	F-TM-001	**Screen**	lib/screens/team_members/team_members_screen.dart
Title	Team members — client-side filter	**Role**	ADMIN / HR
Priority	P2	**Type**	Functional
Pre-conditions	100 team members.		
Screen flow / Steps	**1.** Search 'eng'; filter dept='Engineering'.		
Expected UI behaviour	Client-side filter shows only Engineering matches.		
Construction context	—	**Flutter gap**	F-G18 — no server pagination; all loaded — scale risk.

lib/main.dart

ID	F-SYNC-004	**Screen**	lib/main.dart
Title	Outbox not initialised on web	**Role**	any (web)
Priority	P2	**Type**	Offline Gap
Pre-conditions	Browser.		
Screen flow / Steps	**1.** Open Pending Sync route.		
Expected UI behaviour	Route hidden / placeholder. Today: route shown but reads empty.		
Construction context	—	**Flutter gap**	F-G30 — web variant has no Drift backend.

ID	F-RES-001	**Screen**	lib/main.dart
Title	Global error handler catches uncaught async errors	**Role**	any
Priority	P2	**Type**	Resilience
Pre-conditions	Force a deliberate throw in a screen build.		
Screen flow / Steps	**1.** Trigger.		
Expected UI behaviour	FlutterError.onError + PlatformDispatcher.onError logged; screen shows red Material error box (debug) or generic fallback (release).		
Construction context	—	**Flutter gap**	F-G26 — no per-route ErrorBoundary widget.

ID	F-UPG-002	**Screen**	lib/main.dart
Title	Non-blocking upgrade prompt allows skip	**Role**	any
Priority	P2	**Type**	Compatibility
Pre-conditions	Installed v3.0; latest v3.1 (non-breaking).		
Screen flow / Steps	**1.** Open app.		
Expected UI behaviour	Dismissable banner with 'Update available'. Skippable.		
Construction context	—	**Flutter gap**	—

ID	F-LIFE-003	**Screen**	lib/main.dart
Title	Hot reload retains route but not transient state	**Role**	dev
Priority	P3	**Type**	Resilience
Pre-conditions	Dev build.		
Screen flow / Steps	**1.** Trigger hot reload while on Gantt.		
Expected UI behaviour	Gantt re-renders on same route; CPM provider may re-fetch.		
Construction context	—	**Flutter gap**	—

multiple

ID	F-RES-001	**Screen**	multiple
Title	RESIDENTIAL — full mobile field-ops day-flow	**Role**	SITE_ENGINEER (mobile)
Priority	P1	**Type**	End-to-End
Pre-conditions	Residential project assigned.		
Screen flow / Steps	**1.** Login on mobile. **2.** Drawer → My Tasks → filter Active. **3.** Open task; capture progress 60% with note; Save. **4.** Open Site Reports → New; take 3 geotagged photos; submit. **5.** Switch network off; capture and submit a second report (queued). **6.** Switch network on; observe Pending Sync drain.		
Expected UI behaviour	All actions complete; queued report syncs without duplicate; PM inbox shows mark-completes for approval.		
Construction context	Residential G+2 villa.	**Flutter gap**	—

ID	F-COM-001	**Screen**	multiple
Title	COMMERCIAL — multi-phase Gantt & inter-state PO	**Role**	PM (desktop)
Priority	P1	**Type**	End-to-End
Pre-conditions	Commercial project; vendor in another state.		
Screen flow / Steps	**1.** Apply 'Commercial-Standard-G+4' WBS template. **2.** Add inter-state PO (Maharashtra vendor). **3.** Issue invoice from a stage in Karnataka project.		
Expected UI behaviour	WBS cloned; PO uses IGST (per backend). Invoice should show IGST split — today only single GST line.		
Construction context	Commercial G+4.	**Flutter gap**	F-G06 — IGST split UI missing.

ID	F-INT-001	**Screen**	multiple
Title	INTERIOR — DPC heavy on customisations	**Role**	ESTIMATOR / PM
Priority	P1	**Type**	End-to-End
Pre-conditions	Lead 'interior_work'.		
Screen flow / Steps	**1.** Generate estimation in interior mode. **2.** Convert to project; approve BOQ (no structural items). **3.** Issue DPC with 6 customisations (modular kitchen, wardrobes).		
Expected UI behaviour	Interior template excludes structural scopes; DPC PDF lists customisations in dedicated section.		
Construction context	Interior 3BHK fit-out.	**Flutter gap**	—

ID	F-REN-001	**Screen**	multiple
Title	RENOVATION — pre-work photos + demolition deduction	**Role**	PM
Priority	P1	**Type**	End-to-End
Pre-conditions	Existing house; demolition planned.		
Screen flow / Steps	**1.** Gallery → upload 20 pre-work photos. **2.** BOQ → mark salvage as DEDUCTION item (negative impact). **3.** Issue stage payment schedule 25/25/25/25.		
Expected UI behaviour	Pre-work photos archived; deductions reflected in net contract value.		
Construction context	Renovation with selective demolition.	**Flutter gap**	—

ID	F-SMH-001	**Screen**	multiple
Title	SMART-HOME — add-on CO with OTP customer approval	**Role**	PM
Priority	P2	**Type**	End-to-End
Pre-conditions	Existing project.		
Screen flow / Steps	**1.** Create CO 'Smart home IoT kit' ■3.5L. **2.** Submit → Internal Approve → Send to customer. **3.** Customer enters OTP; approves. **4.** Start work; commissioning QC check.		
Expected UI behaviour	Status → IN_PROGRESS; advance invoice generated; warranty row added on completion.		
Construction context	Smart-home retrofit.	**Flutter gap**	—

ID	F-IND-006	**Screen**	multiple
Title	Date format DD/MM/YYYY across screens	**Role**	any
Priority	P3	**Type**	Localisation
Pre-conditions	Device en_IN.		
Screen flow / Steps	**1.** Open Task Create, Site Report list, BOQ doc.		
Expected UI behaviour	Dates consistently DD MMM YYYY or DD/MM/YYYY.		
Construction context	—	**Flutter gap**	F-G16 — some date strings use literal `{day}/{month}/{year}` not locale-aware.

ID	F-OFF-004	**Screen**	multiple
Title	Time-zone drift between device and server	**Role**	any
Priority	P2	**Type**	Edge
Pre-conditions	Device clock set 10 mins ahead.		
Screen flow / Steps	**1.** Submit a site report.		
Expected UI behaviour	Server stores UTC of actual receipt; UI later displays in device local (IST). No date display jump.		
Construction context	—	**Flutter gap**	—

ID	F-PROV-001	**Screen**	multiple
Title	Provider disposed on screen pop avoids leaks	**Role**	dev / QA
Priority	P2	**Type**	Memory
Pre-conditions	Open / close Gantt 50 times.		
Screen flow / Steps	**1.** Trigger heap snapshot.		
Expected UI behaviour	GanttCpmProvider, CpmTaskResult lists garbage-collected; no ever-growing retain.		
Construction context	—	**Flutter gap**	F-G18 — audit Provider lifecycle in heavy screens.

Finance / vendor payment dialog

ID	F-IND-003	**Screen**	Finance / vendor payment dialog
Title	TDS field on vendor payment — gap	**Role**	FINANCE
Priority	P1	**Type**	India Compliance Gap
Pre-conditions	Vendor invoice.		
Screen flow / Steps	**1.** Record payment.		
Expected UI behaviour	TDS section selector (194C/194J), rate, slip download. Today: absent.		
Construction context	—	**Flutter gap**	F-G07 — no TDS UI.

Project create / customer projects

ID	F-IND-007	**Screen**	Project create / customer projects
Title	RERA number on residential project (>8 units)	**Role**	PM
Priority	P1	**Type**	India Compliance Gap
Pre-conditions	Residential project >8 units.		
Screen flow / Steps	**1.** Open project create form.		
Expected UI behaviour	Should show RERA registration number field. Today: missing.		
Construction context	—	**Flutter gap**	F-G06 — no RERA capture (ties to backend G-27).

theme / lib/theme

ID	F-A11Y-002	**Screen**	theme / lib/theme/app_theme.dart
Title	Contrast on coral red brand colour	**Role**	any
Priority	P3	**Type**	Accessibility
Pre-conditions	High-contrast device.		
Screen flow / Steps	**1.** Measure contrast of coralRed buttons on white.		
Expected UI behaviour	WCAG AA >= 4.5; coralRed (#F36F72) on white likely fails — needs darker variant for text-on-white.		
Construction context	—	**Flutter gap**	F-G29 — verify WCAG conformance.

lib/theme/app_theme.dart

ID	F-THM-001	**Screen**	lib/theme/app_theme.dart
Title	Dark mode toggle absent	**Role**	any
Priority	P3	**Type**	Theme Gap
Pre-conditions	Anywhere.		
Screen flow / Steps	**1.** Look for dark mode toggle in profile / settings.		
Expected UI behaviour	Today: not implemented.		
Construction context	—	**Flutter gap**	F-G27 — light theme only.

all

ID	F-LOC-001	**Screen**	all
Title	Hindi / Kannada strings — not present	**Role**	any
Priority	P3	**Type**	Localisation Gap
Pre-conditions	Device locale hi_IN or kn_IN.		
Screen flow / Steps	**1.** Open any screen.		
Expected UI behaviour	All strings English. Today: no `AppLocalizations` or .arb files.		
Construction context	—	**Flutter gap**	F-G28 — no i18n.

ID	F-SEC-002	**Screen**	all
Title	Logs do not contain auth token	**Role**	any
Priority	P1	**Type**	Security
Pre-conditions	Profile screen.		
Screen flow / Steps	**1.** Inspect console / logcat during navigation.		
Expected UI behaviour	No 'Bearer ' string appears in logs.		
Construction context	—	**Flutter gap**	—

lib/services/outbox_service.dart

ID	F-OFF-003	**Screen**	lib/services/outbox_service.dart
Title	Outbox photo file deleted by OS cleanup	**Role**	SITE_ENGINEER
Priority	P2	**Type**	Resilience
Pre-conditions	Photo cached in tmp; OS deletes due to space pressure.		
Screen flow / Steps	**1.** Open Pending Sync; retry.		
Expected UI behaviour	Service detects missing file; row moves to permanent failure with clear reason.		
Construction context	—	**Flutter gap**	—

ID	F-IDM-001	**Screen**	lib/services/outbox_service.dart
Title	Outbox attaches X-Idempotency-Key on financial mutations	**Role**	any
Priority	P1	**Type**	Functional
Pre-conditions	Offline; user marks stage certified; outbox enqueues.		
Screen flow / Steps	**1.** Reconnect; outbox drains. **2.** Force a retry of the same entry.		
Expected UI behaviour	Each outbox row carries a stable UUID v4 idempotency key generated at enqueue time. Retry uses same key. Backend dedupes; ledger shows exactly one effect.		
Construction context	—	**Flutter gap**	F-G22 — must cover stage certify, payment confirm, invoice raise.

lib/services/api_service.dart

ID	F-RES-003	**Screen**	lib/services/api_service.dart
Title	401 mid-action triggers silent refresh	**Role**	any
Priority	P1	**Type**	Resilience
Pre-conditions	Access token expired during a form submit.		
Screen flow / Steps	**1.** Submit form.		
Expected UI behaviour	ApiService interceptor calls /auth/refresh-token; retries original request once; user sees nothing.		
Construction context	—	**Flutter gap**	—

lib/services/storage_service.dart

ID	F-SEC-003	**Screen**	lib/services/storage_service.dart
Title	Storage service writes secure on mobile	**Role**	any
Priority	P1	**Type**	Security
Pre-conditions	Mobile.		
Screen flow / Steps	**1.** Login; inspect.		
Expected UI behaviour	Tokens via flutter_secure_storage (Keychain / Keystore). Plain SharedPreferences only on web (web has limitations).		
Construction context	—	**Flutter gap**	—

lib/main.dart + lib/services

ID	F-PUSH-001	**Screen**	lib/main.dart + lib/services/push_notification_service.dart
Title	Tap FCM push deep-links to the right screen	**Role**	any
Priority	P1	**Type**	Functional
Pre-conditions	App backgrounded; server sends STAGE_CERTIFIED push with {type:'stage', stageId:42, projectId:7}.		
Screen flow / Steps	**1.** Tap the notification on the lock screen.		
Expected UI behaviour	App resumes and navigates to PaymentScheduleScreen of project 7 with stage 42 scrolled into view. Today: routing may stop at default landing.		
Construction context	—	**Flutter gap**	F-G31 — verify FCM payload → GoRouter resolution.

lib/services/push_notification_service.dart

ID	F-PUSH-002	**Screen**	lib/services/push_notification_service.dart
Title	Foreground push surfaces in-app banner not OS notification	**Role**	any
Priority	P2	**Type**	UX
Pre-conditions	App open on Leads screen.		
Screen flow / Steps	**1.** Receive LEAD_ASSIGNED push.		
Expected UI behaviour	In-app banner / Snackbar 'New lead assigned'; tap routes to lead. No duplicate OS notif.		
Construction context	—	**Flutter gap**	—

ID	F-FCM-DL-001	**Screen**	lib/services/push_notification_service.dart
Title	STAGE_DUE push deep-links to PaymentSchedule with stage scrolled-into-view	**Role**	any
Priority	P1	**Type**	Functional
Pre-conditions	Backgrounded; server emits payload {type:'STAGE_DUE', stageId:42, projectId:7}.		
Screen flow / Steps	**1.** Tap notification.		
Expected UI behaviour	App resumes; navigates to PaymentScheduleScreen of project 7; stage 42 row is scrolled into view and visually highlighted for 2s. If deep-link fails, falls back to landing with toast.		
Construction context	—	**Flutter gap**	F-G31 — confirm route table covers STAGE_DUE, INVOICE_RAISED, LEAD_ASSIGNED, FA_AGREED.

ID	F-FCM-DL-002	**Screen**	lib/services/push_notification_service.dart
Title	Two pushes within window — only one in-app banner	**Role**	any
Priority	P2	**Type**	UX
Pre-conditions	App foregrounded.		
Screen flow / Steps	**1.** Server emits two STAGE_DUE for same stage within 1 min.		
Expected UI behaviour	Client de-dupes by stageId+kind+yyyymmdd; only one banner; subsequent suppressed (mirrors server INSERT-ON-CONFLICT dedup).		
Construction context	—	**Flutter gap**	—

lib/services/portal_auth_service.dart

ID	F-PUSH-003	**Screen**	lib/services/portal_auth_service.dart
Title	FCM token deregistered on logout	**Role**	any
Priority	P1	**Type**	Privacy
Pre-conditions	Logged in with FCM token.		
Screen flow / Steps	**1.** Logout.		
Expected UI behaviour	POST /auth/fcm-token with empty token or DELETE call clears server-side token so subsequent pushes don't reach this device.		
Construction context	—	**Flutter gap**	F-G18 — confirm token clear on logout.

lib/main.dart + version check

ID	F-UPG-001	**Screen**	lib/main.dart + version check
Title	Forced-upgrade screen blocks navigation	**Role**	any
Priority	P1	**Type**	Compatibility
Pre-conditions	Installed app v2.3; minimum supported v3.0.		
Screen flow / Steps	**1.** Open app; observe.		
Expected UI behaviour	Modal 'Update required' with store link. No way to dismiss; cannot reach login.		
Construction context	—	**Flutter gap**	F-G18 — confirm min-version policy is wired.

customer-facing routes (TBD)

ID	F-CST-001	**Screen**	customer-facing routes (TBD)
Title	Customer portal — project list scoped to user	**Role**	CUSTOMER
Priority	P1	**Type**	RBAC
Pre-conditions	Customer with 2 projects.		
Screen flow / Steps	**1.** Login on /customer.		
Expected UI behaviour	Only the 2 projects visible; navigation rail / drawer restricted to Documents, BOQ Acknowledge, Payments, Support.		
Construction context	—	**Flutter gap**	F-G06 — customer portal coverage minimal in this codebase (majority is internal staff app).

customer BOQ document view

ID	F-CST-002	**Screen**	customer BOQ document view
Title	Customer acknowledges BOQ from portal	**Role**	CUSTOMER
Priority	P1	**Type**	Functional
Pre-conditions	BOQ APPROVED internally.		
Screen flow / Steps	**1.** Customer opens BOQ on portal; taps Acknowledge.		
Expected UI behaviour	customerAcknowledgedAt/By stored; UI shows 'Acknowledged on ' chip; no status change.		
Construction context	—	**Flutter gap**	—

customer payment screen

ID	F-CST-003	**Screen**	customer payment screen
Title	Customer pays stage via UPI (deep link)	**Role**	CUSTOMER
Priority	P2	**Type**	India / Payment
Pre-conditions	Stage DUE.		
Screen flow / Steps	**1.** Customer taps 'Pay via UPI'.		
Expected UI behaviour	upi:// intent fires with payee VPA + amount. After payment, user returns; receipt awaits manual confirmation by Finance.		
Construction context	—	**Flutter gap**	F-G06 — UPI deep-link integration not visible in repo.

forms across app

ID	F-KBD-002	**Screen**	forms across app
Title	Done action on phone keyboard advances to next field	**Role**	any
Priority	P3	**Type**	UX
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Tap 'Next' on numeric keyboard.		
Expected UI behaviour	Focus moves to next TextFormField rather than dismissing keyboard prematurely.		
Construction context	—	**Flutter gap**	F-G18 — many forms omit textInputAction = TextInputAction.next.

any notes field

ID	F-LONG-002	**Screen**	any notes field
Title	Notes field accepts multiline (newlines preserved)	**Role**	any
Priority	P3	**Type**	Functional
Pre-conditions	Form open.		
Screen flow / Steps	**1.** Type 'Line1\nLine2' (Shift+Enter on web).		
Expected UI behaviour	Newlines preserved in display.		
Construction context	—	**Flutter gap**	—

web build

ID	F-WEB-001	**Screen**	web build
Title	Web build PWA install / service worker	**Role**	any
Priority	P3	**Type**	Web
Pre-conditions	Chrome 120+.		
Screen flow / Steps	**1.** Open portal in browser.		
Expected UI behaviour	Install prompt available (PWA manifest); service worker caches static assets.		
Construction context	—	**Flutter gap**	F-G30 — service worker offline strategy missing.

lead documents / brochure (BrochureController API)

ID	F-BRO-001	**Screen**	lead documents / brochure (BrochureController API)
Title	Brochure download from lead page	**Role**	SALES
Priority	P3	**Type**	Functional
Pre-conditions	Brochures uploaded by admin.		
Screen flow / Steps	**1.** Open lead; select brochure to share.		
Expected UI behaviour	PDF downloads or share-sheet opens (WhatsApp, email).		
Construction context	—	**Flutter gap**	F-G24 — brochure download UI presence not verified.

lib/screens/boq

ID	F-HSN-001	**Screen**	lib/screens/boq/boq_line_editor.dart
Title	HSN/SAC visible + editable on BOQ line row	**Role**	ESTIMATOR
Priority	P1	**Type**	Functional
Pre-conditions	BOQ line list view.		
Screen flow / Steps	**1.** Open a line; inspect form.		
Expected UI behaviour	HSN/SAC text field with input formatter restricting to 4–8 digits. Pre-filled from selected material's hsn_sac_code (V137). Save fails with 'invalid HSN' if regex violates.		
Construction context	—	**Flutter gap**	F-G37 — HSN now shown on row chip too.

ID	F-AGREED-001	**Screen**	lib/screens/boq/boq_line_editor.dart
Title	AGREED BOQ line shows lock icon; rate field disabled	**Role**	ESTIMATOR
Priority	P1	**Type**	Business Rule
Pre-conditions	Line state=AGREED; stage-1 invoice raised.		
Screen flow / Steps	**1.** Try to edit unit rate.		
Expected UI behaviour	Rate field disabled with tooltip 'Locked after invoice — raise a Change Order'. CO button appears.		
Construction context	—	**Flutter gap**	—

ID	F-RESP-001	**Screen**	lib/screens/boq/boq_list_screen.dart
Title	BOQ table → cards transition at < 600px	**Role**	any
Priority	P2	**Type**	Responsive
Pre-conditions	DevTools at 1200px then 360px.		
Screen flow / Steps	**1.** Resize.		
Expected UI behaviour	DataTable layout on wide. Card list with horizontal-scroll chips on narrow. No clipped totals; no horizontal overflow indicator.		
Construction context	—	**Flutter gap**	—

lib/screens/final_account

ID	F-409-001	**Screen**	lib/screens/final_account/final_account_editor.dart
Title	Final Account stale edit shows 409 conflict UX	**Role**	PROJECT_MANAGER
Priority	P1	**Type**	Concurrency
Pre-conditions	User A and B both have FA open; B saves first.		
Screen flow / Steps	**1.** User A taps Save.		
Expected UI behaviour	Banner 'Final Account changed by another user' with two buttons: 'Discard and reload' (refetch + replace local form) and 'Keep my edits' (download fresh + show inline diff). No silent overwrite.		
Construction context	—	**Flutter gap**	F-G23 — wires backend 409 to a usable client flow.

lib/screens/customers

ID	F-CST-LC-001	**Screen**	lib/screens/customers/customer_list_screen.dart
Title	Customer kebab shows Activate/Deactivate, no Delete (regress 1aa1fb46)	**Role**	ADMIN
Priority	P1	**Type**	Refactor regression
Pre-conditions	Customer list.		
Screen flow / Steps	**1.** Open row kebab.		
Expected UI behaviour	Only Activate or Deactivate is shown (mutex on state). Delete is absent from kebab. Delete reachable only from a separate confirm flow (7e04b70d).		
Construction context	—	**Flutter gap**	—

ID	F-CST-LC-002	**Screen**	lib/screens/customers/customer_list_screen.dart
Title	Server outcome surfaced — Deleted vs Deactivated toast (regress ad2632d9)	**Role**	ADMIN
Priority	P1	**Type**	UX
Pre-conditions	Customer has linked lead.		
Screen flow / Steps	**1.** Perform delete on a deletable customer. **2.** Perform delete on a customer with linked lead.		
Expected UI behaviour	Toast text differs: 'Customer deleted' vs 'Customer deactivated (has linked records)'. List row removed or recoloured accordingly.		
Construction context	—	**Flutter gap**	—

lib/services/outbox_service.dart + Drift schema

ID	F-IDM-002	**Screen**	lib/services/outbox_service.dart + Drift schema
Title	Idempotency key persists across app kill / device reboot	**Role**	any
Priority	P1	**Type**	Reliability
Pre-conditions	Outbox row with key 'k1' pending.		
Screen flow / Steps	**1.** Kill app; reboot device; open app.		
Expected UI behaviour	Row still has 'k1'; drain uses same key. Drift schema persists the column; key never regenerated on resume.		
Construction context	—	**Flutter gap**	—

lib/utils/money_format.dart (or shared)

ID	F-MONEY-001	**Screen**	lib/utils/money_format.dart (or shared)
Title	Money fields render as Indian lakh/crore with 2dp	**Role**	any
Priority	P1	**Type**	Localisation
Pre-conditions	Amount 12,345,678.5 paise.		
Screen flow / Steps	**1.** Render in invoice, BOQ summary, dashboard tile.		
Expected UI behaviour	Display '■1,23,45,678.51' (Indian grouping). HALF_UP at 2dp matches server MoneyMath. Negative values prefixed with '-' not '()'.		
Construction context	—	**Flutter gap**	F-G28 — shared Indian numeral formatter still missing in places.

lib/screens/invoice

ID	F-MONEY-002	**Screen**	lib/screens/invoice/invoice_preview_screen.dart
Title	Cumulative GST roll-up matches server within 1 paise	**Role**	FINANCE
Priority	P1	**Type**	Regression
Pre-conditions	50-line invoice in preview.		
Screen flow / Steps	**1.** Compare on-screen total vs server-issued PDF.		
Expected UI behaviour	Difference ≤ ■0.01 on both subtotal and GST.		
Construction context	—	**Flutter gap**	—

lib/screens/leads

ID	F-LEAD-INT-001	**Screen**	lib/screens/leads/lead_detail_screen.dart
Title	Lead detail shows interactions timeline + next-follow-up chip	**Role**	SALES
Priority	P1	**Type**	Functional
Pre-conditions	Lead with 3 interactions: CALL, WHATSAPP, EMAIL.		
Screen flow / Steps	**1.** Open lead.		
Expected UI behaviour	Timeline ordered desc by occurred_at; channel icon + outcome + notes preview. Next-follow-up chip turns red when overdue. Add Interaction CTA visible.		
Construction context	—	**Flutter gap**	—

ID	F-LEAD-INT-002	**Screen**	lib/screens/leads/quick_log_interaction_sheet.dart
Title	Quick-log Interaction sheet — minimal taps to record CALL outcome	**Role**	SALES
Priority	P2	**Type**	UX
Pre-conditions	Lead detail open.		
Screen flow / Steps	**1.** Tap Quick-log → CALL → No answer → Save.		
Expected UI behaviour	Bottom sheet with channel chips, outcome chips, optional notes, next-follow-up date picker. Total ≤ 3 taps for the common case.		
Construction context	—	**Flutter gap**	—

lib/app_shell.dart + offline gates

ID	F-WEB-001	**Screen**	lib/app_shell.dart + offline gates
Title	Web build hides offline-only entry points	**Role**	any
Priority	P2	**Type**	UX / Web
Pre-conditions	Build kIsWeb=true.		
Screen flow / Steps	**1.** Inspect drawer + dashboard.		
Expected UI behaviour	Pending Sync, My Tasks (offline mode), Outbox screen are hidden on web (F-G30). A small disclaimer 'Offline mode unavailable on web — use mobile app for field work' shown in profile.		
Construction context	—	**Flutter gap**	—

7. Flutter implementation gaps register

Each gap below was found while mapping the Flutter codebase for this catalogue and is referenced from the matching test cases via the **Flutter gap** cell. Backend-side gaps remain documented in the companion *PORTAL_TEST_CASES.pdf*.

Gap ID	Area	Description
F-G01	Auth	Login regex outdated (TLD <= 4 chars); rejects valid emails like .museum.
F-G02	Auth	Reset-token guard is only length>20, not UUID/sig format check.
F-G03	Routing	Mixed Navigator.push and go_router; no deep-link for /projects/:id.
F-G04	Gantt	Non-virtualised chart list; performance risk at 100+ tasks on mid-range devices.
F-G05	Tasks	No client-side predecessor cycle detection; relies on backend response.
F-G06	India / GST	No CGST/SGST/IGST split UI; no RERA capture; no place-of-supply selector.
F-G07	India / TDS	No TDS section selector, rate input, or TDS slip download anywhere.
F-G08	Materials / HSN	No HSN/SAC input on material form (backend G-21 added column; UI not wired).
F-G09	CCTV	Camera password field stored plaintext in form; no auth test pre-save.
F-G10	ProjectDetail	Permission-gated tabs (schedule-config) hidden silently without 'Access Denied' affordance.
F-G11	Subcontract / Approval	Detail screens stubbed — show snackbar TODO.
F-G12	Tasks / Monsoon	Monsoon-sensitive flag stored but absent from task list / create / detail; only Gantt warning chip.
F-G13	PM Approval Inbox	Photo only 56 px thumbnail; no fullscreen preview; no bulk approve.
F-G14	WBS Templates	Predecessor editor UI missing (only count shown); duration always 'days'; no hours/weeks selector.
F-G15	Holiday Calendar	No project-specific override flag; cannot disable a state holiday for a single project.
F-G16	Localisation / Dates	Some screens use literal `{day}/{month}/{year}`; not locale-aware.
F-G17	State Management	No PageStorageKey on long lists; scroll position lost; menu expansion state ad-hoc; active visit banner not global.
F-G18	UX general	Stale caches on re-entry (role list, partner lookup); no draft auto-save; silent background-refresh failures; partial doc upload progress; no GST auto-calc on VO; no approver workflow on stock adjust; no PF/ESI in wage sheet.
F-G19	Search / Lists	No debounce on search bars; type counts computed only from current page; no infinite-scroll guard against duplicate fetch; chip list not auto-scrolled to selection; bulk attendance has no RFID/Aadhaar capture.
F-G20	Forms	Multi-step wizards have no progress indicator and no auto-save.
F-G21	Confirmations	Missing on irreversible actions: BOQ approve, DPC Issue, stage certify, FA agree, retention release.

F-G22	Idempotency	No X-Idempotency-Key on stage certify, payment confirm, invoice raise; network retry could duplicate.
F-G23	Concurrency	No optimistic-lock 409 conflict UI for FinalAccount (backend G-19 fix needs client follow-through).
F-G24	PDF Preview / Export	No in-app PDF preview before invoice/certificate issue; no PDF for VO; no print/share/annotate in viewer.
F-G25	View 360	No VR mode, pinch-zoom dependent on `panorama` pkg internals.
F-G26	Error handling	No per-route ErrorBoundary widget; reliance on global handlers only.
F-G27	Theme	Light theme only; no dark-mode toggle, no font-scale setting.
F-G28	Localisation	No i18n (`AppLocalizations` / .arb absent); no shared Indian lakh/crore formatter.
F-G29	Accessibility / Privacy	Few Semantics widgets; coral brand colour likely fails WCAG AA contrast for buttons; no client-side Aadhaar masking helper.
F-G30	Offline / Web	OutboxService + drift skipped on web — web users have no offline capability for site reports / task mark-complete.
F-G31	Notifications	No type filter; no delete / archive; badge not refreshed on app resume.
F-G32	Support / SLA	No SLA badge in list; no attachments in reply UI; no bulk actions.
F-G33	Warranties / DLP	No DLP countdown chip; no expiry alert notifications.
F-G34	Documents	Legacy documents screen has delete-only actions; no upload progress indicator; no Office-doc preview.
F-G35	Portal Users	Role picker shows all roles; not project-scoped.
F-G36	Password change	After /reset-password no client-side relogin; user can still call APIs until token refresh fails.
F-G37	BOQ list	No GST split per item; no HSN/SAC display; subcontract retention not visible per row.
F-G38	Subcontract list	Retention only in detail; missing from card row in list view.
F-G39	GPS	No mock-location detection client-side for site reports / visits.
F-G40	Evidence	No client-side EXIF inspection / time-jump plausibility check.
F-G44	BOQ Editor	HSN/SAC inherit-on-material-select not wired in editor; estimator has to retype.
F-G45	Concurrency UX	Backend 409 on Final Account, AGREED rate, and BOQ approve lacks dedicated client banner / diff view in places.
F-G46	Project Lifecycle	Admin delete typed-confirmation dialog not enforced; risk of accidental tap-through on sensitive action.
F-G47	Customer Lifecycle	Toast text not differentiated for Delete vs Deactivate outcomes when server returns 200 in both cases.
F-G48	Idempotency	Outbox idempotency-key persistence across reboot relies on Drift column not yet confirmed for every mutation type.

F-G49	Push routing	STAGE_DUE / INVOICE_RAISED / FA_AGREED / LEAD_ASSIGNED deep-link table not exhaustive; fallback may surface as no-op.
F-G50	Indian numerals	[CLOSED] All NumberFormat.currency(symbol:'■') call sites now pass locale:'en_IN' — lakh/crore grouping consistent across 11 screens (BOQ, FA, deductions, payment schedule, VO, stage payments, project detail, etc.).

8. Construction-type traceability

Construction type	Key Flutter test cases
Residential	F-RES-001, F-LEAD-003, F-TSK-PROG-001, F-SR-001, F-LAB-001, F-STG-002
Commercial	F-COM-001, F-WBS-003, F-PRC-001, F-BOQ-006, F-IND-002
Interior	F-INT-001, F-LEAD-009, F-DPC-001, F-DPC-003
Renovation	F-REN-001, F-GAL-001, F-DEL-001
Vastu	F-LEAD-003
Smart Home	F-SMH-001, F-BOQ-005, F-WAR-001

9. Priority distribution

Priority	Count	Definition
P1	89	Blocker — financial, security, compliance or data-loss critical
P2	109	High — major functional / workflow / construction-domain breakage
P3	43	Medium — UX, localisation, peripheral feature